DRoberts Case Study 1 DS6306

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Establish library calls

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(ggplot2)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(tidyr)
library(knitr)
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
```

```
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(class)
library(e1071)
```

```
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```r
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
library(magick)
```

```
## Linking to ImageMagick 6.9.13.29
## Enabled features: cairo, fontconfig, freetype, heic, lcms, pango, raw, rsvg, webp
## Disabled features: fftw, ghostscript, x11
```

Ingest data for evaluation

```r
employee_data = read.csv("/Users/dianewroberts/Case Study 1/CaseStudy1-data.csv", sep = ",", header = TR

head(employee_data)
```

```
##   ID Age Attrition    BusinessTravel DailyRate             Department
## 1  1  32        No     Travel_Rarely       117                  Sales
## 2  2  40        No     Travel_Rarely      1308 Research & Development
## 3  3  35        No Travel_Frequently       200 Research & Development
## 4  4  32        No     Travel_Rarely       801                  Sales
## 5  5  24        No Travel_Frequently       567 Research & Development
## 6  6  27        No Travel_Frequently       294 Research & Development
##   DistanceFromHome Education   EducationField EmployeeCount EmployeeNumber
## 1               13         4    Life Sciences             1            859
## 2               14         3          Medical             1           1128
## 3               18         2    Life Sciences             1           1412
## 4                1         4        Marketing             1           2016
## 5                2         1 Technical Degree             1           1646
## 6               10         2    Life Sciences             1            733
##   EnvironmentSatisfaction Gender HourlyRate JobInvolvement JobLevel
## 1                       2   Male         73              3        2
## 2                       3   Male         44              2        5
## 3                       3   Male         60              3        3
## 4                       3 Female         48              3        3
```

```
## 5                       1 Female         32              3        1
## 6                       4   Male         32              3        3
##                  JobRole JobSatisfaction MaritalStatus MonthlyIncome
## 1        Sales Executive               4      Divorced          4403
## 2      Research Director               3        Single         19626
## 3 Manufacturing Director               4        Single          9362
## 4        Sales Executive               4       Married         10422
## 5     Research Scientist               4        Single          3760
## 6 Manufacturing Director               1      Divorced          8793
##   MonthlyRate NumCompaniesWorked Over18 OverTime PercentSalaryHike
## 1        9250                  2      Y       No                11
## 2       17544                  1      Y       No                14
## 3       19944                  2      Y       No                11
## 4       24032                  1      Y       No                19
## 5       17218                  1      Y      Yes                13
## 6        4809                  1      Y       No                21
##   PerformanceRating RelationshipSatisfaction StandardHours StockOptionLevel
## 1                 3                        3            80               1
## 2                 3                        1            80               0
## 3                 3                        3            80               0
## 4                 3                        3            80               2
## 5                 3                        3            80               0
## 6                 4                        3            80               2
##   TotalWorkingYears TrainingTimesLastYear WorkLifeBalance YearsAtCompany
## 1                 8                     3               2             5
## 2                21                     2               4            20
## 3                10                     2               3             2
## 4                14                     3               3            14
## 5                 6                     2               3             6
## 6                 9                     4               2             9
##   YearsInCurrentRole YearsSinceLastPromotion YearsWithCurrManager
## 1                  2                       0                    3
## 2                  7                       4                    9
## 3                  2                       2                    2
## 4                 10                       5                    7
## 5                  3                       1                    3
## 6                  7                       1                    7
```

```r
str(employee_data)
```

```
## 'data.frame':    870 obs. of  36 variables:
##  $ ID                     : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Age                    : int  32 40 35 32 24 27 41 37 34 34 ...
##  $ Attrition              : chr  "No" "No" "No" "No" ...
##  $ BusinessTravel         : chr  "Travel_Rarely" "Travel_Rarely" "Travel_Frequently" "Travel_Rarely"
##  $ DailyRate              : int  117 1308 200 801 567 294 1283 309 1333 653 ...
##  $ Department             : chr  "Sales" "Research & Development" "Research & Development" "Sales"
##  $ DistanceFromHome       : int  13 14 18 1 2 10 5 10 10 10 ...
##  $ Education              : int  4 3 2 4 1 2 5 4 4 4 ...
##  $ EducationField         : chr  "Life Sciences" "Medical" "Life Sciences" "Marketing" ...
##  $ EmployeeCount          : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ EmployeeNumber         : int  859 1128 1412 2016 1646 733 1448 1105 1055 1597 ...
##  $ EnvironmentSatisfaction : int  2 3 3 3 1 4 2 4 3 4 ...
##  $ Gender                 : chr  "Male" "Male" "Male" "Female" ...
##  $ HourlyRate             : int  73 44 60 48 32 32 90 88 87 92 ...
```

```
##  $ JobInvolvement          : int  3 2 3 3 3 3 4 2 3 2 ...
##  $ JobLevel                : int  2 5 3 3 1 3 1 2 1 2 ...
##  $ JobRole                 : chr  "Sales Executive" "Research Director" "Manufacturing Director" "Sal
##  $ JobSatisfaction         : int  4 3 4 4 4 1 3 4 3 3 ...
##  $ MaritalStatus           : chr  "Divorced" "Single" "Single" "Married" ...
##  $ MonthlyIncome           : int  4403 19626 9362 10422 3760 8793 2127 6694 2220 5063 ...
##  $ MonthlyRate             : int  9250 17544 19944 24032 17218 4809 5561 24223 18410 15332 ...
##  $ NumCompaniesWorked      : int  2 1 2 1 1 1 2 2 1 1 ...
##  $ Over18                  : chr  "Y" "Y" "Y" "Y" ...
##  $ OverTime                : chr  "No" "No" "No" "No" ...
##  $ PercentSalaryHike       : int  11 14 11 19 13 21 12 14 19 14 ...
##  $ PerformanceRating       : int  3 3 3 3 3 4 3 3 3 3 ...
##  $ RelationshipSatisfaction: int  3 1 3 3 3 3 1 3 4 2 ...
##  $ StandardHours           : int  80 80 80 80 80 80 80 80 80 80 ...
##  $ StockOptionLevel        : int  1 0 0 2 0 2 0 3 1 1 ...
##  $ TotalWorkingYears       : int  8 21 10 14 6 9 7 8 1 8 ...
##  $ TrainingTimesLastYear   : int  3 2 2 3 2 4 5 5 2 3 ...
##  $ WorkLifeBalance         : int  2 4 3 3 3 2 2 3 3 2 ...
##  $ YearsAtCompany          : int  5 20 2 14 6 9 4 1 1 8 ...
##  $ YearsInCurrentRole      : int  2 7 2 10 3 7 2 0 1 2 ...
##  $ YearsSinceLastPromotion : int  0 4 2 5 1 1 0 0 0 7 ...
##  $ YearsWithCurrManager    : int  3 9 2 7 3 7 3 0 0 7 ...
```

```r
names(employee_data)
```

```
##  [1] "ID"                       "Age"
##  [3] "Attrition"                "BusinessTravel"
##  [5] "DailyRate"                "Department"
##  [7] "DistanceFromHome"         "Education"
##  [9] "EducationField"           "EmployeeCount"
## [11] "EmployeeNumber"           "EnvironmentSatisfaction"
## [13] "Gender"                   "HourlyRate"
## [15] "JobInvolvement"           "JobLevel"
## [17] "JobRole"                  "JobSatisfaction"
## [19] "MaritalStatus"            "MonthlyIncome"
## [21] "MonthlyRate"              "NumCompaniesWorked"
## [23] "Over18"                   "OverTime"
## [25] "PercentSalaryHike"        "PerformanceRating"
## [27] "RelationshipSatisfaction" "StandardHours"
## [29] "StockOptionLevel"         "TotalWorkingYears"
## [31] "TrainingTimesLastYear"    "WorkLifeBalance"
## [33] "YearsAtCompany"           "YearsInCurrentRole"
## [35] "YearsSinceLastPromotion"  "YearsWithCurrManager"
```

Data wrangling - Impute missing fields and ensure that data is categoriezed as needed

```r
#Evaluate if there are blank or NA fields
employee_data %>%
  summarise(across(everything(),
                   ~sum(is.na(.) | . == "", na.rm = TRUE)))
```

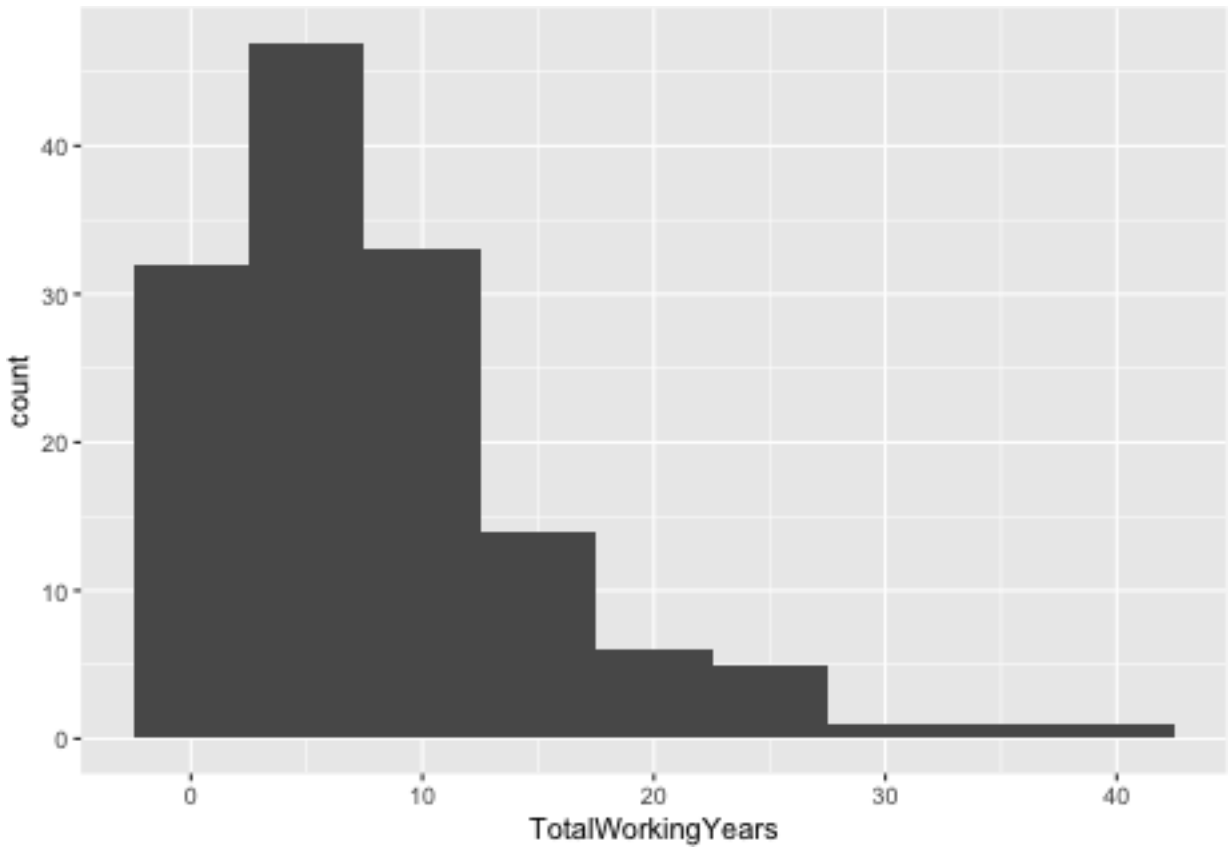
```
##   ID Age Attrition BusinessTravel DailyRate Department DistanceFromHome
## 1  0   0         0               0         0          0                 0
##   Education EducationField EmployeeCount EmployeeNumber EnvironmentSatisfaction
```

```
## 1         0              0             0              0                        0
##   Gender HourlyRate JobInvolvement JobLevel JobRole JobSatisfaction
## 1      0          0              0        0       0               0
##   MaritalStatus MonthlyIncome MonthlyRate NumCompaniesWorked Over18 OverTime
## 1             0             0           0                  0      0        0
##   PercentSalaryHike PerformanceRating RelationshipSatisfaction StandardHours
## 1                 0                 0                        0             0
##   StockOptionLevel TotalWorkingYears TrainingTimesLastYear WorkLifeBalance
## 1                0                 0                     0               0
##   YearsAtCompany YearsInCurrentRole YearsSinceLastPromotion
## 1              0                  0                       0
##   YearsWithCurrManager
## 1                    0
```

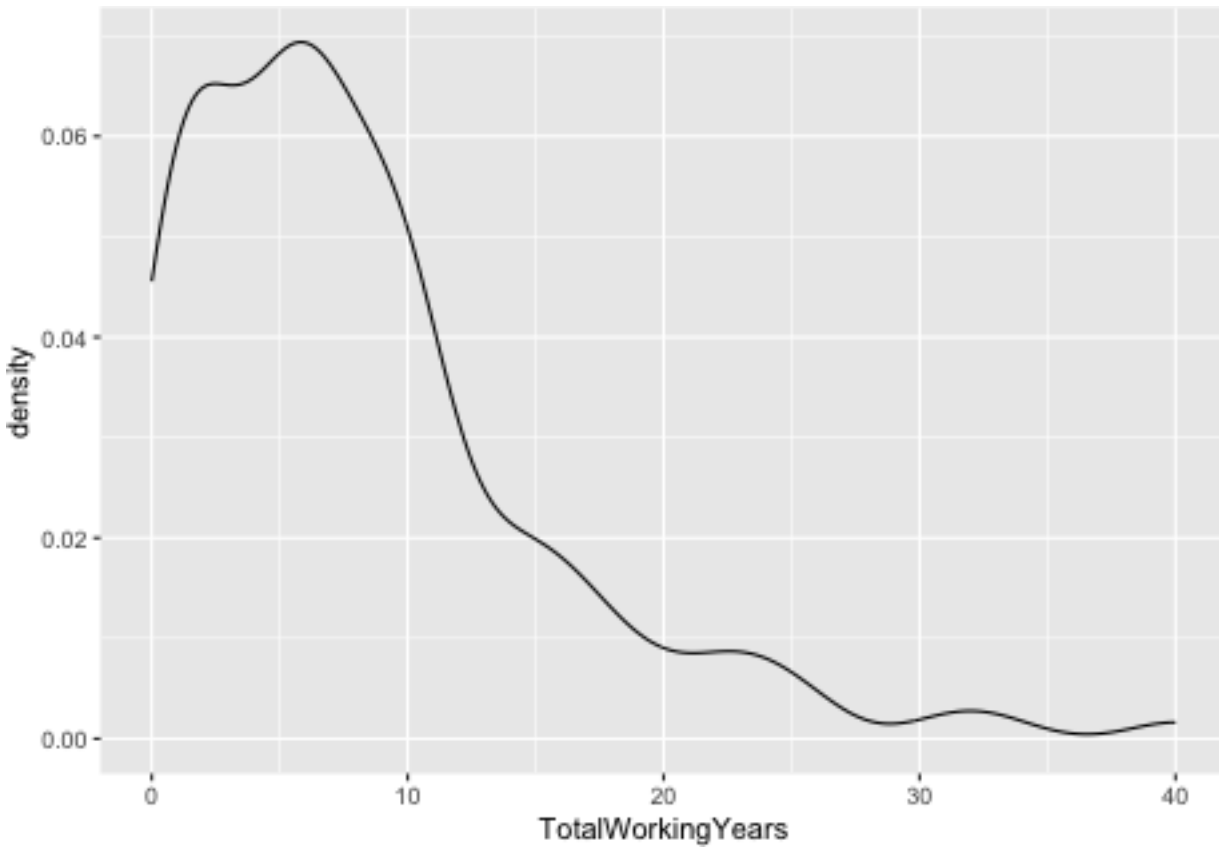
```
#No blanks or NA fields located

employee_data_clean <- employee_data %>%
  mutate(Attrition_factor = as.factor(Attrition),
         BusinessTravel_factor = as.factor(BusinessTravel),
         Department_factor = as.factor(Department),
         Education_factor = as.factor(Education),
         EducationField_factor = as.factor(EducationField),
         EnvironmentSatisfaction_factor = as.factor(EnvironmentSatisfaction),
         Gender_factor = as.factor(Gender),
         JobInvolvement_factor = as.factor(JobInvolvement),
         JobLevel_factor = as.factor(JobLevel),
         JobSatisfaction_factor = as.factor(JobSatisfaction),
         MaritalStatus_factor = as.factor(MaritalStatus),
         OverTime_factor = as.factor(OverTime),
         PerformanceRating_factor = as.factor(PerformanceRating),
         RelationshipSatisfaction_factor = as.factor(RelationshipSatisfaction),
         StockOptionLevel_factor = as.factor(StockOptionLevel),
         WorkLifeBalance_factor = as.factor(WorkLifeBalance))

view(employee_data_clean)

# Create a dataset that only has employees who have left the company for visualization purposes

employee_attrition = employee_data_clean %>%
  filter(Attrition_factor == "Yes")

view(employee_attrition)
```

Visualization of data

```
## In 5 year increments determine the highest number of employees attriting by the total number of years

ggplot(employee_attrition, aes(x = TotalWorkingYears)) +
  geom_histogram(binwidth = 5)
```



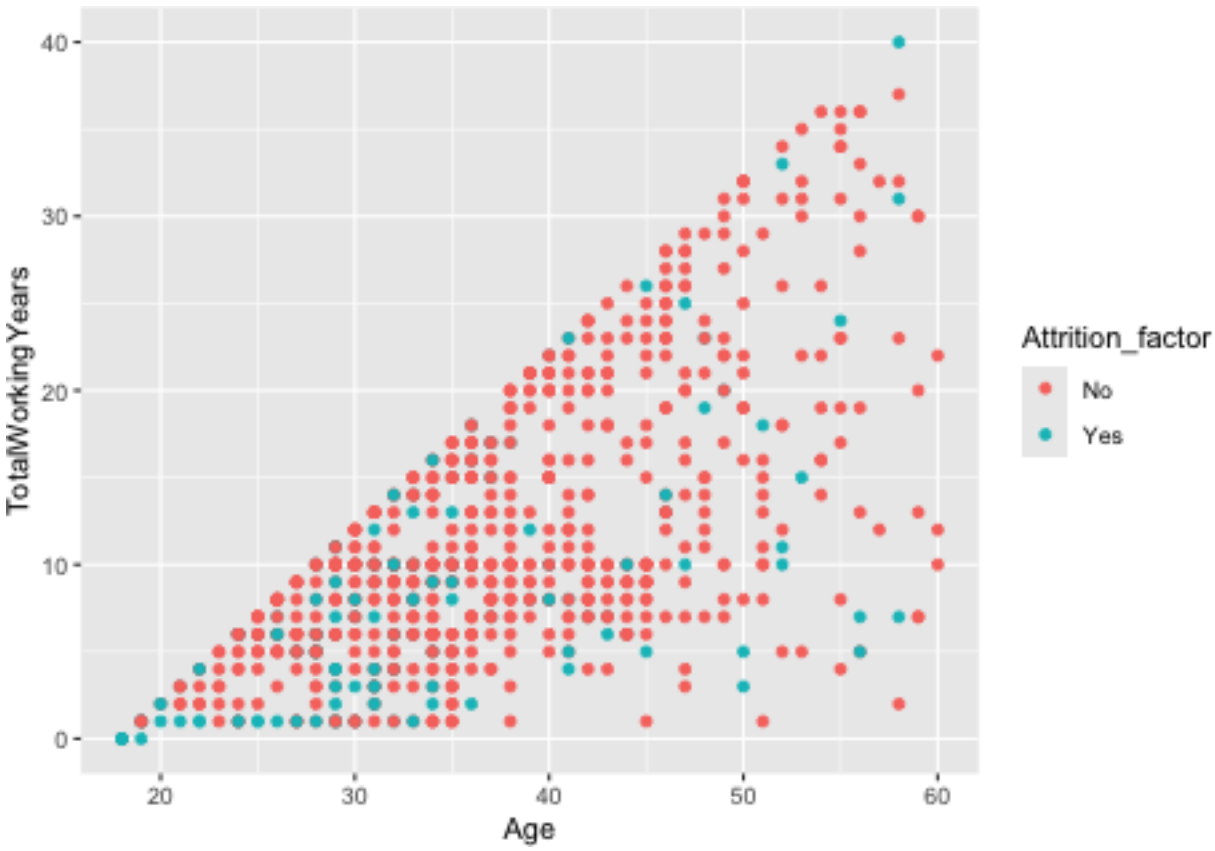

```
# The histogram appears to indicate that employees are most likely to attrit at <12 years and there is

# To view this smoothly I will create a density plot

ggplot(employee_attrition, aes(x = TotalWorkingYears)) +
  geom_density()
```



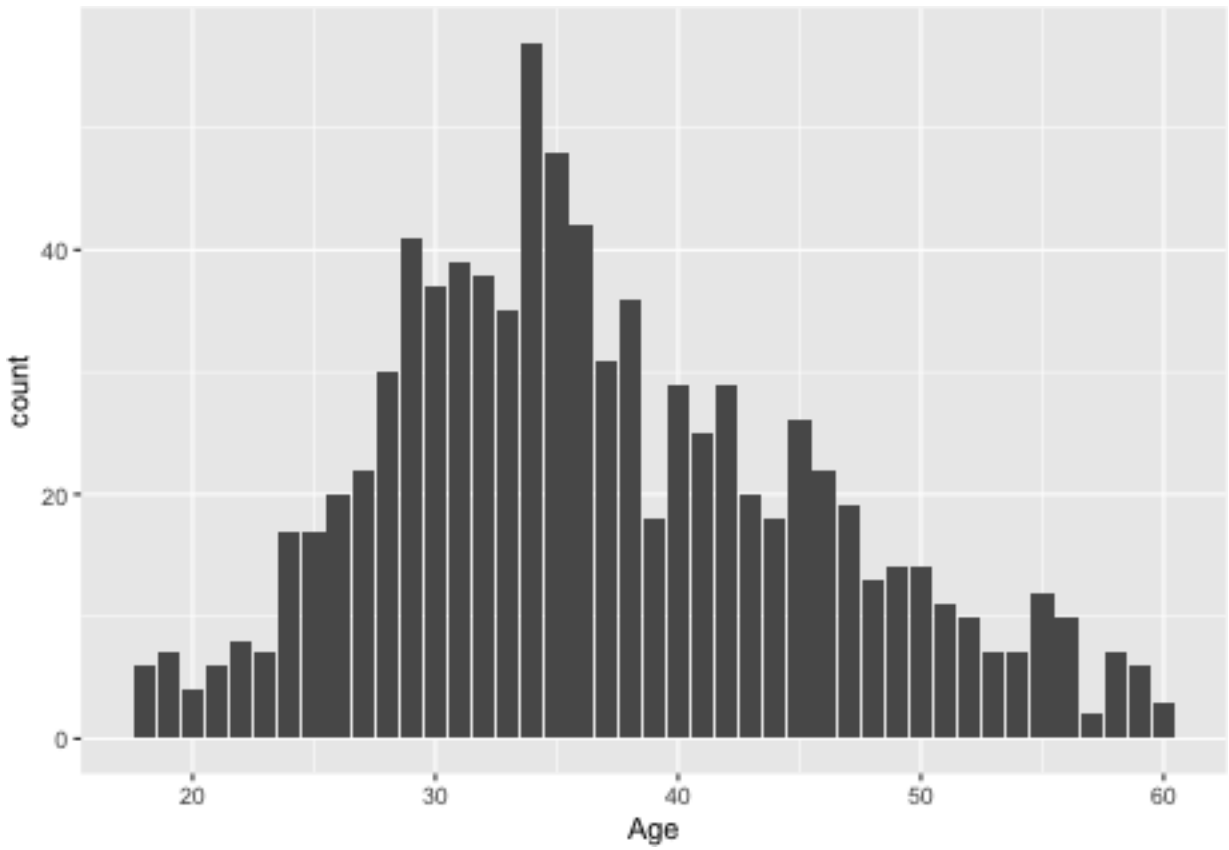

```
## Creating a dataset with those employees who have not attrited

#Original plot, worked to create an exec slide-friendly version
#employee_notattrition = employee_data_clean %>%
#  filter(Attrition_factor == "No")
#ggplot(employee_notattrition, aes(x = TotalWorkingYears)) +
#  geom_density()

cutoff <- 12

employee_notattrition <- employee_data_clean %>%
  filter(Attrition_factor == "No")

ggplot(employee_notattrition, aes(x = TotalWorkingYears)) +

  # Base density curve
  geom_density(fill = "gray85",
               color = "gray20",
               linewidth = 1.2) +

  # Highlight region under same density curve
  geom_density(
    aes(fill = after_stat(ifelse(x < cutoff, "highlight", "normal"))),
    alpha = 0.9,
    color = NA
  ) +
```

```r
  scale_fill_manual(values = c("highlight" = "darkorange",
                               "normal" = NA),
                    guide = "none") +

  geom_vline(xintercept = cutoff,
             linetype = "dashed",
             linewidth = 1) +

  labs(
    title = "Total Working Years Distribution",
    x = "Total Working Years",
    y = "Density"
  ) +

  theme_minimal(base_size = 18) +
  theme(
    panel.grid.minor = element_blank(),
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(color = "black"),   # show numbers
    axis.ticks.y = element_line(color = "black")   # show ticks
  )
```

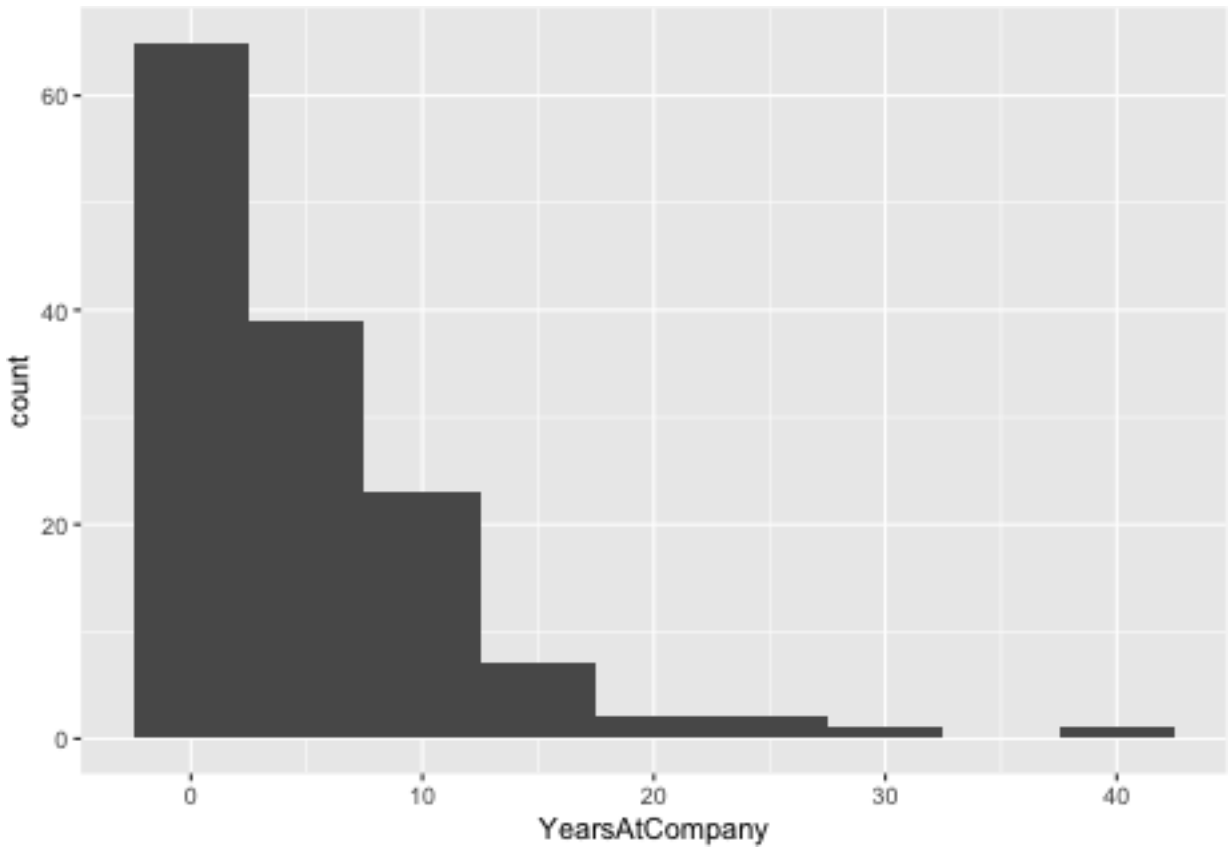
```r
# Conduct a plot across all employees to determine age in years and highlight by color the difference w

ggplot(employee_data_clean, aes(x = Age, y = TotalWorkingYears)) +
  geom_point(mapping = aes(color = Attrition_factor))
```




```r
ggplot(employee_data_clean, aes(x = Age)) +
  geom_bar()
```




```
## In 5 year increments determine the highest number of employees attriting by the number of years work

ggplot(employee_attrition, aes(x = YearsAtCompany)) +
  geom_histogram(binwidth = 5)
```



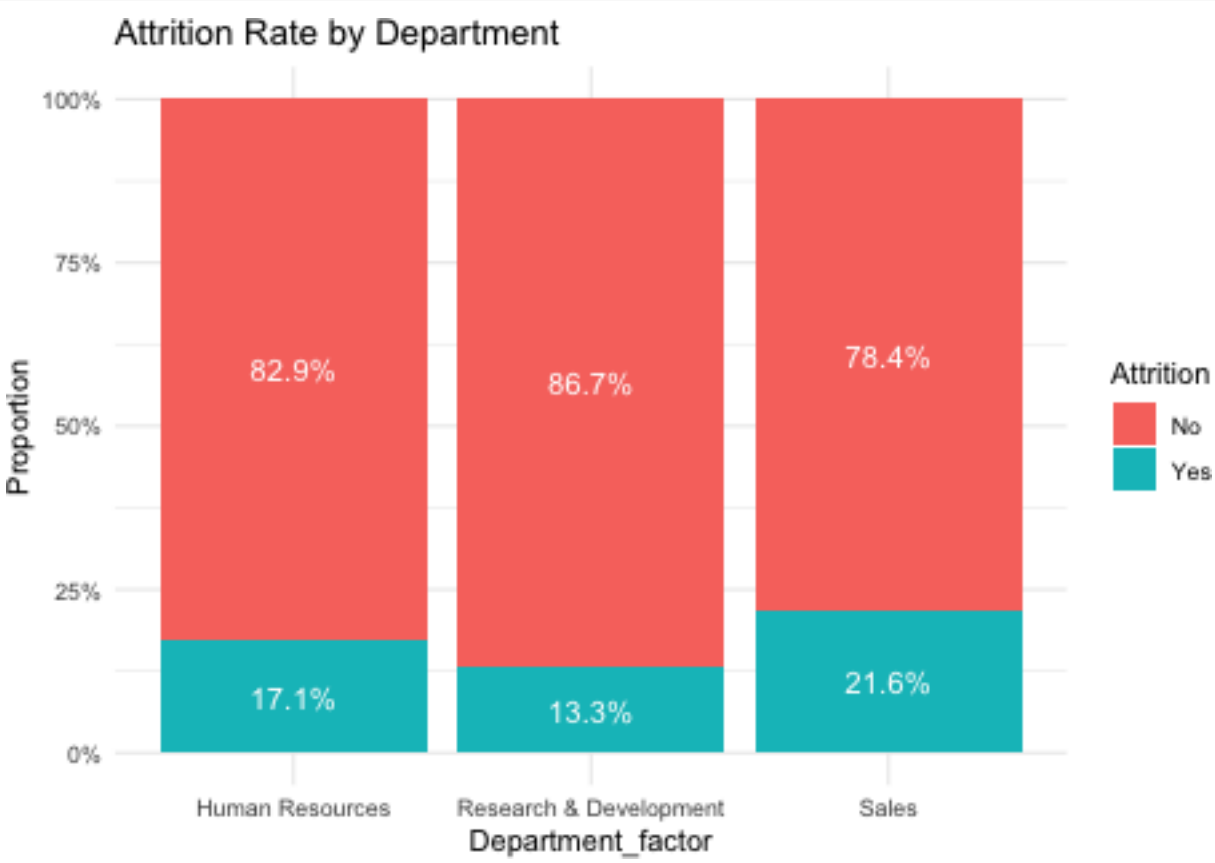

```r
## Now to evaluate the Attrition rate by Department

pivot_table <- employee_data_clean %>%
  count(Department, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Department",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Department")
```

Table 1: Attrition Rate by Department

Department	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Human Resources	29	6	35	17.1%
Research & Development	487	75	562	13.3%

Department	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Sales	214	59	273	21.6%

```r
plot_df <- employee_data_clean %>%
  count(Department_factor, Attrition_factor, name = "n") %>%
  group_by(Department_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df, aes(x = Department_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Department",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



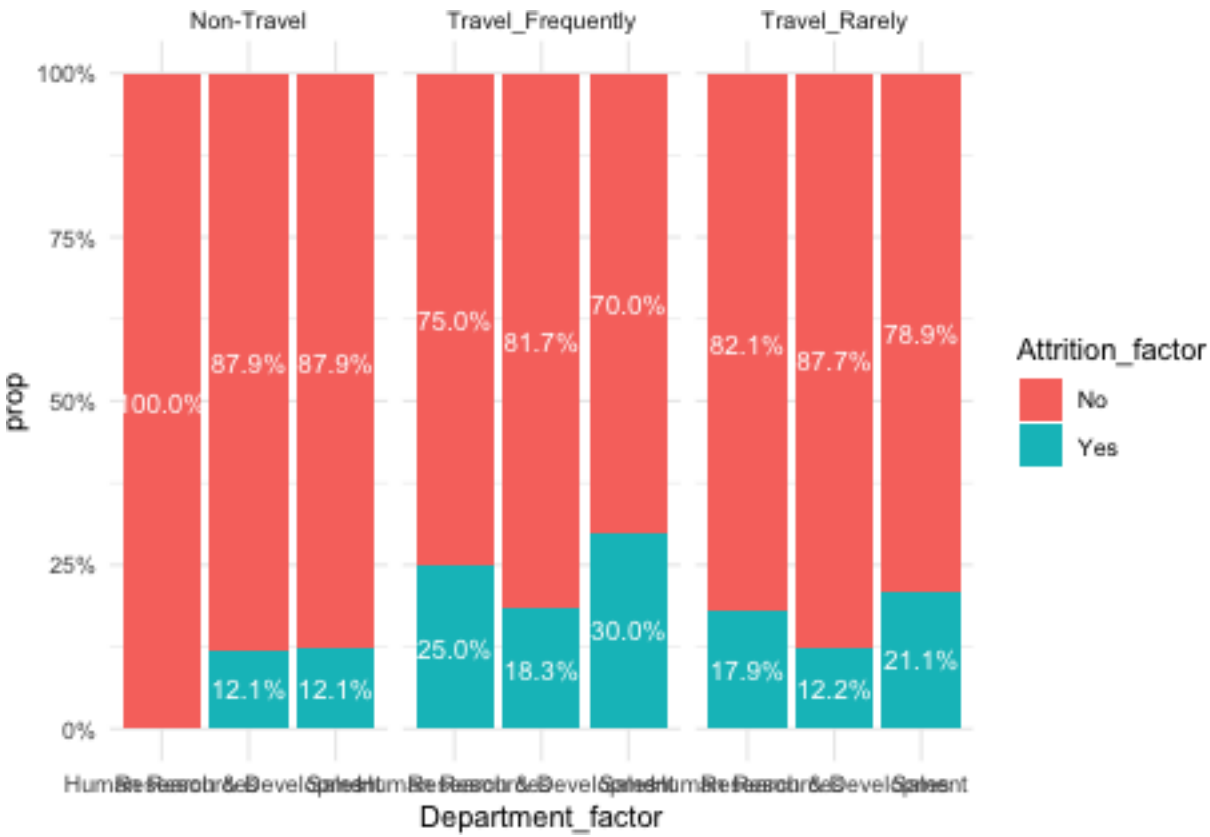

```r
# Do a comparison by Department for Business Travel and attrition
```

```r
pivot_table <- employee_data_clean %>%
  count(BusinessTravel, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Business Travel",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Business Travel")
```

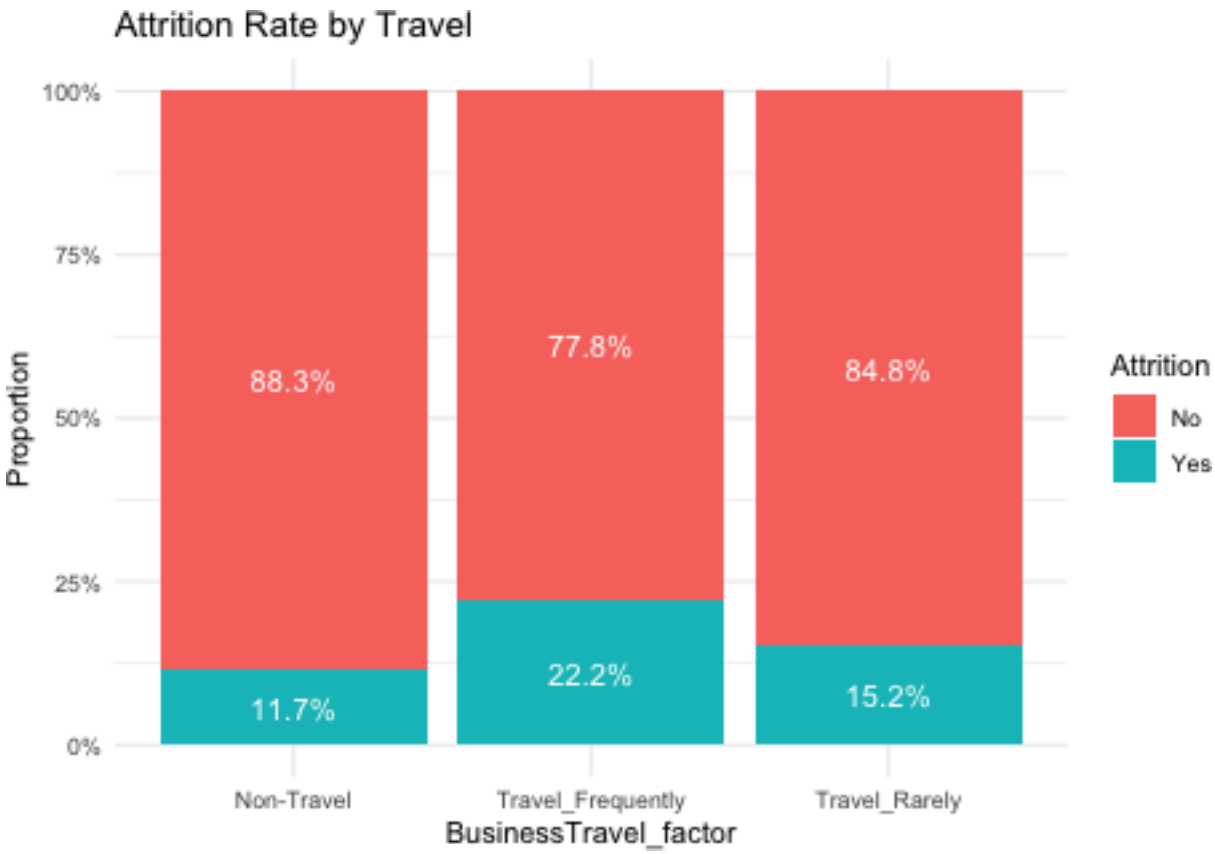
Table 2: Attrition Rate by Business Travel

Business Travel	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Non-Travel	83	11	94	11.7%
Travel_Frequently	123	35	158	22.2%
Travel_Rarely	524	94	618	15.2%

```r
plot_df <- employee_data_clean %>%
  count(Department_factor, BusinessTravel_factor, Attrition_factor, name = "n") %>%
  group_by(Department_factor, BusinessTravel_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = ifelse(prop > 0.05, percent(prop, accuracy = 0.1), "")
  ) %>%
  ungroup()

ggplot(plot_df,
       aes(x = Department_factor,
           y = prop,
           fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 3.5) +
  facet_wrap(~ BusinessTravel_factor) +
  scale_y_continuous(labels = percent_format()) +
  theme_minimal()
```

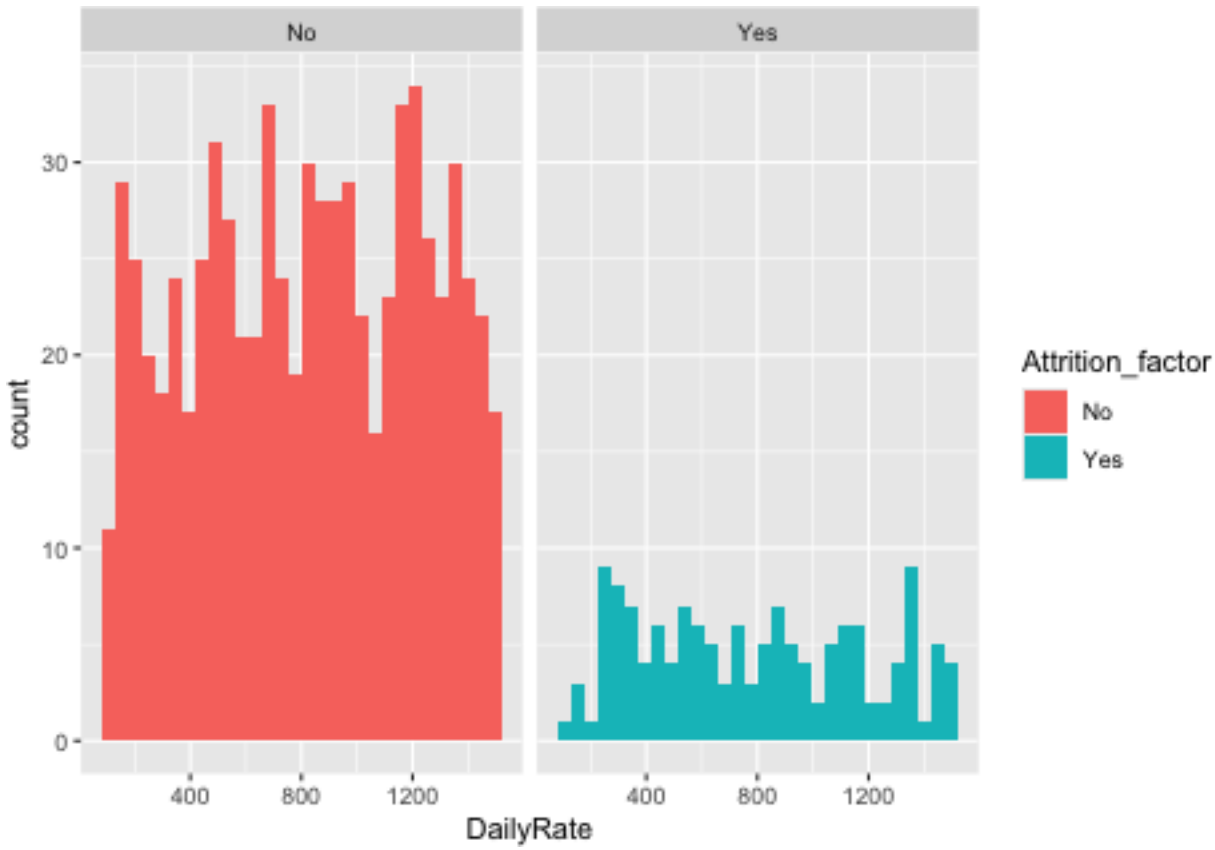



```
# Now I would like to visualize the impact of travel on overall attrition; categorical x categorical

plot_df_3 <- employee_data_clean %>%
  count(BusinessTravel_factor, Attrition_factor, name = "n") %>%
  group_by(BusinessTravel_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_3, aes(x = BusinessTravel_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Travel",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



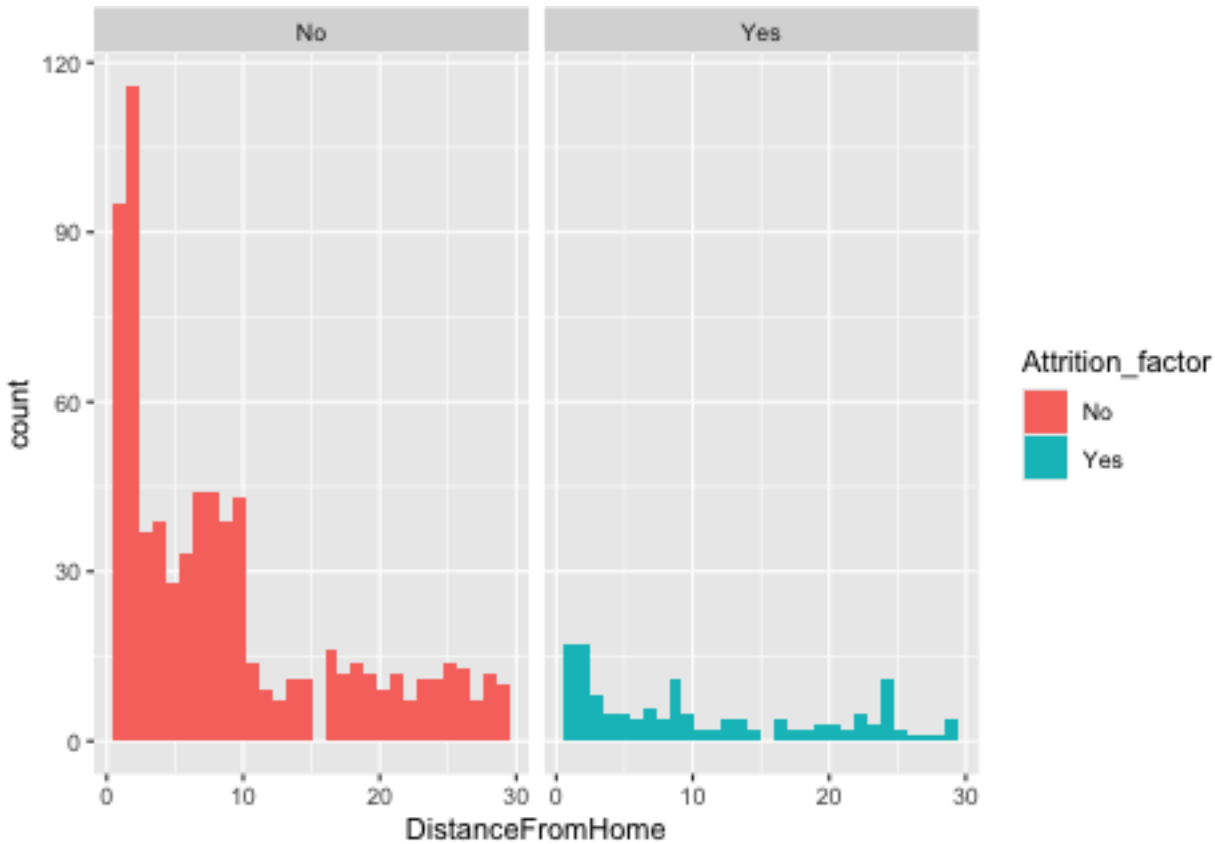

```r
## Next review Daily Salary Rate against Attrition
ggplot(employee_data_clean, aes(x = DailyRate, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
## Distance from home
ggplot(employee_data_clean, aes(x = DistanceFromHome, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



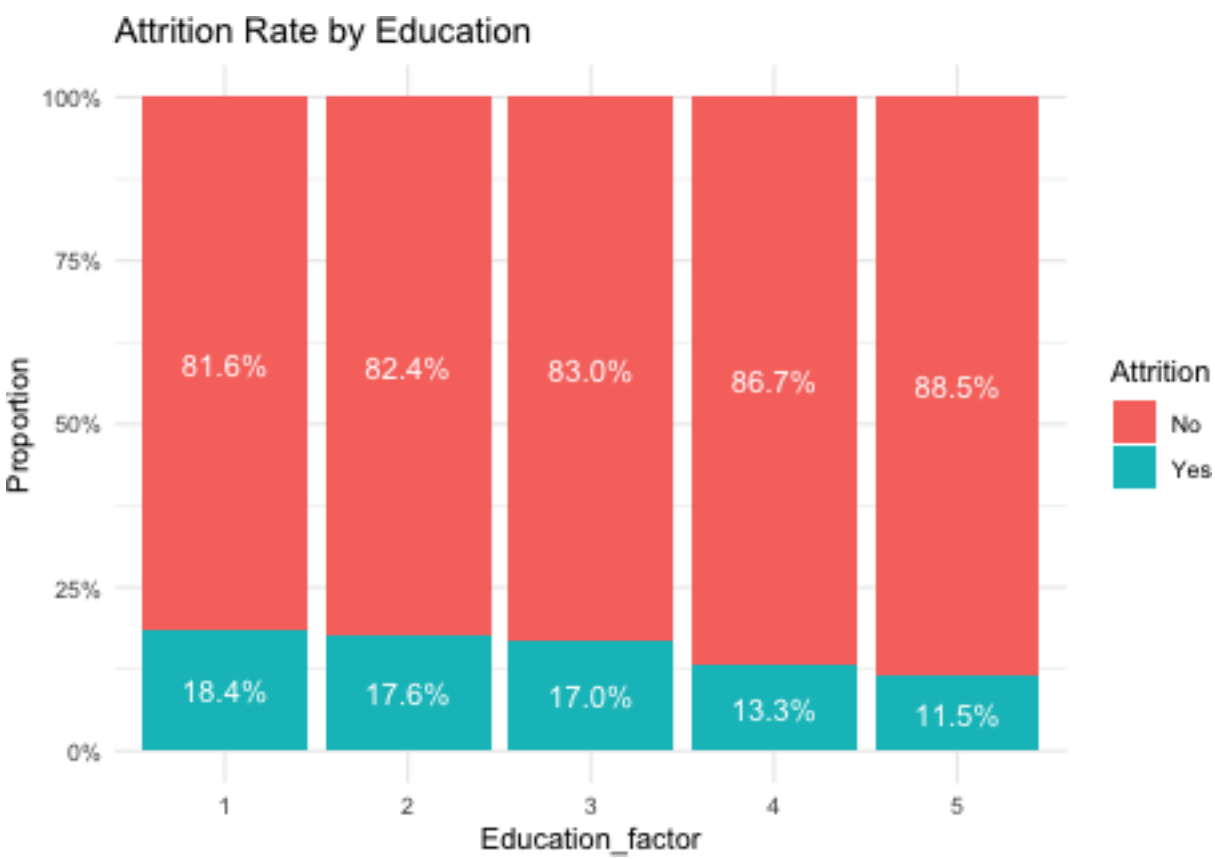

```r
## Education

pivot_table <- employee_data_clean %>%
  count(Education, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Education",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Education")
```

Table 3: Attrition Rate by Education

Education	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	80	18	98	18.4%
2	150	32	182	17.6%
3	269	55	324	17.0%

Education	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
4	208	32	240	13.3%
5	23	3	26	11.5%

```r
plot_df_4 <- employee_data_clean %>%
  count(Education_factor, Attrition_factor, name = "n") %>%
  group_by(Education_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_4, aes(x = Education_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Education",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```

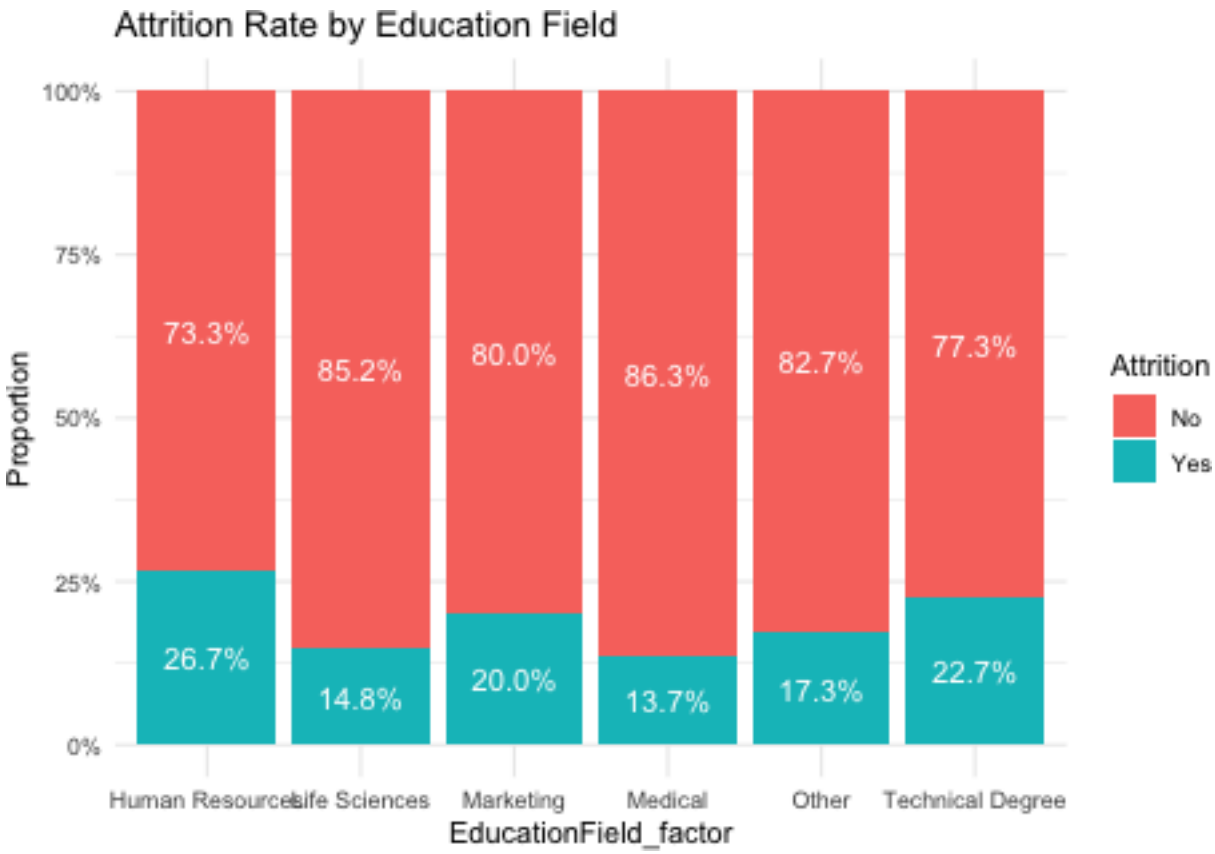


```r
## Education field

pivot_table <- employee_data_clean %>%
  count(EducationField, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Education Field",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Education Field")
```

Table 4: Attrition Rate by Education Field

Education Field	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Human Resources	11	4	15	26.7%
Life Sciences	305	53	358	14.8%
Marketing	80	20	100	20.0%
Medical	233	37	270	13.7%
Other	43	9	52	17.3%
Technical Degree	58	17	75	22.7%

```r
plot_df_5 <- employee_data_clean %>%
  count(EducationField_factor, Attrition_factor, name = "n") %>%
  group_by(EducationField_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_5, aes(x = EducationField_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Education Field",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



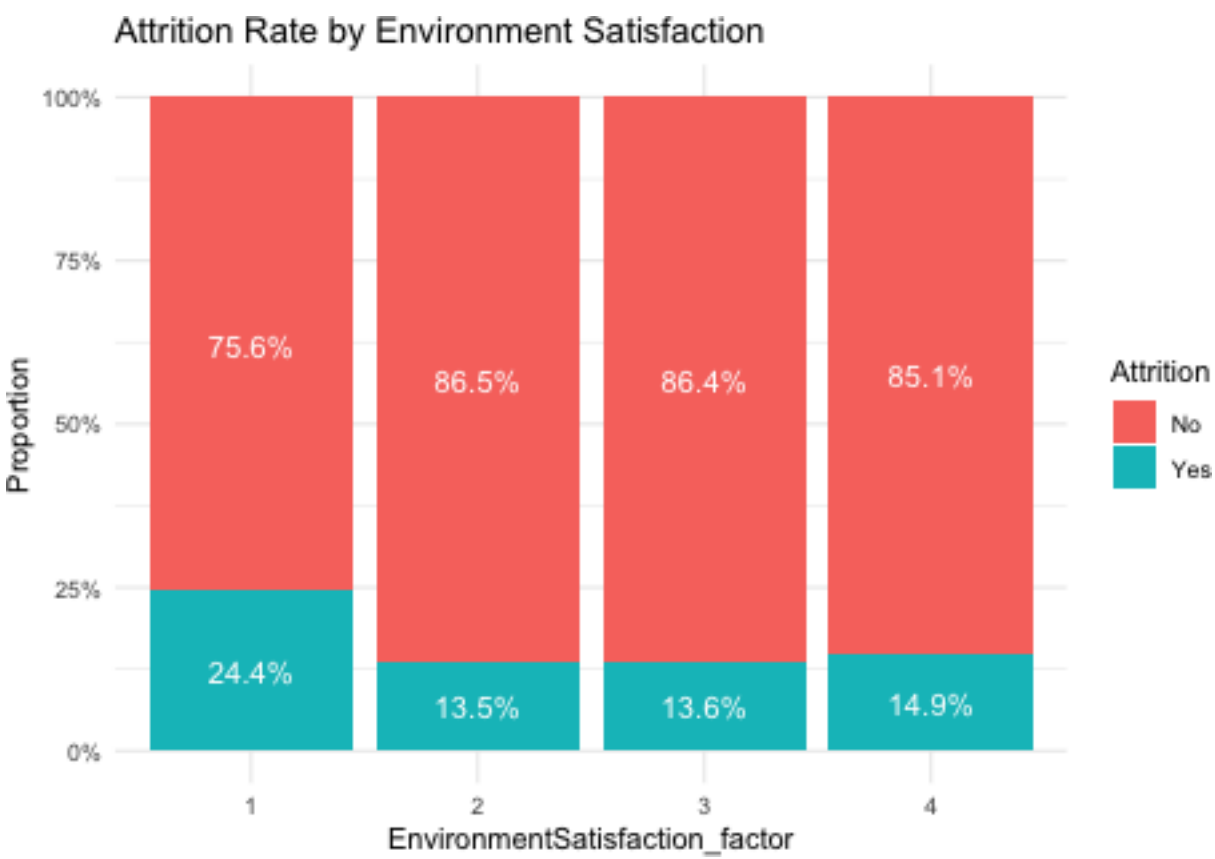

```r
## EnvironmentSatisfaction

pivot_table <- employee_data_clean %>%
  count(EnvironmentSatisfaction, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Environment Satisfaction",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Environment Satisfaction")
```

Table 5: Attrition Rate by Environment Satisfaction

Environment Satisfaction	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	130	42	172	24.4%
2	154	24	178	13.5%

Environment Satisfaction	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
3	223	35	258	13.6%
4	223	39	262	14.9%

```r
plot_df_6 <- employee_data_clean %>%
  count(EnvironmentSatisfaction_factor, Attrition_factor, name = "n") %>%
  group_by(EnvironmentSatisfaction_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_6, aes(x = EnvironmentSatisfaction_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Environment Satisfaction",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```

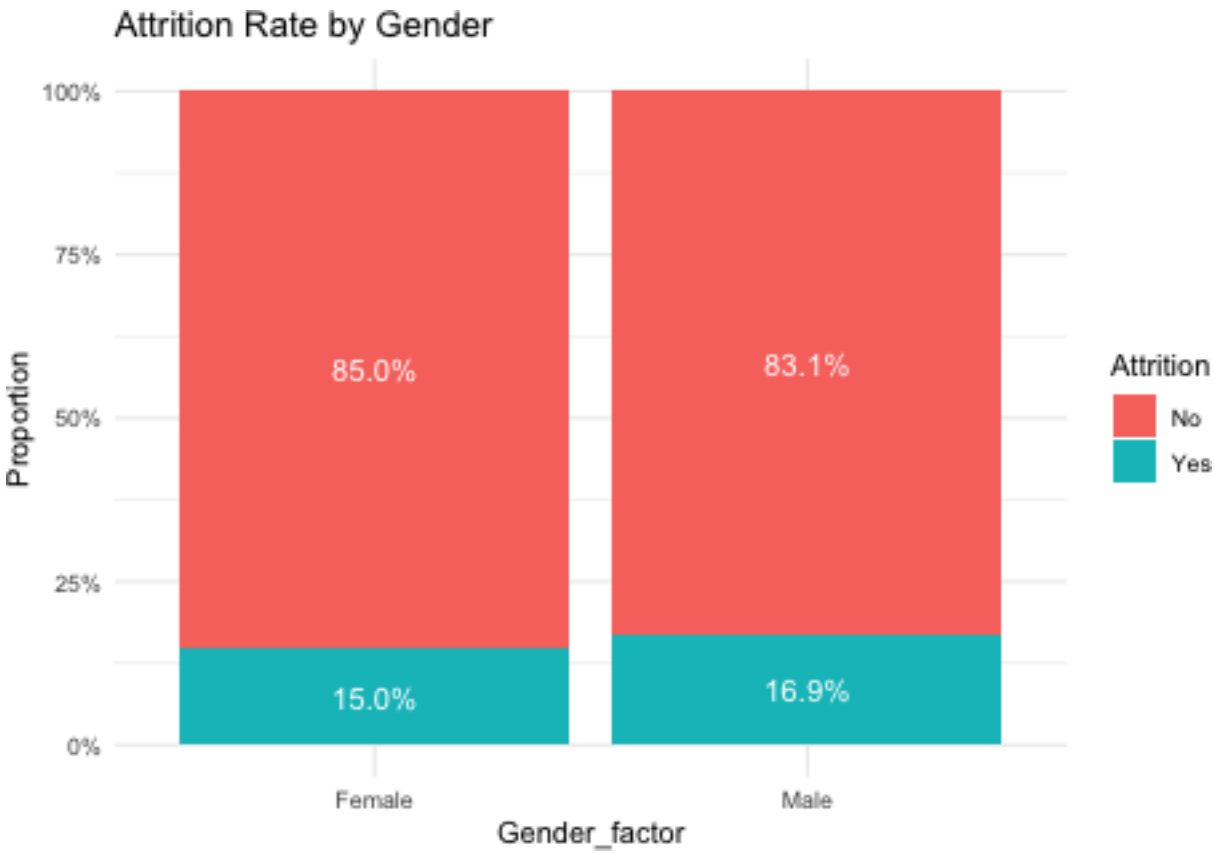


```r
#Gender

pivot_table <- employee_data_clean %>%
  count(Gender, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Gender",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Gender")
```

Table 6: Attrition Rate by Gender

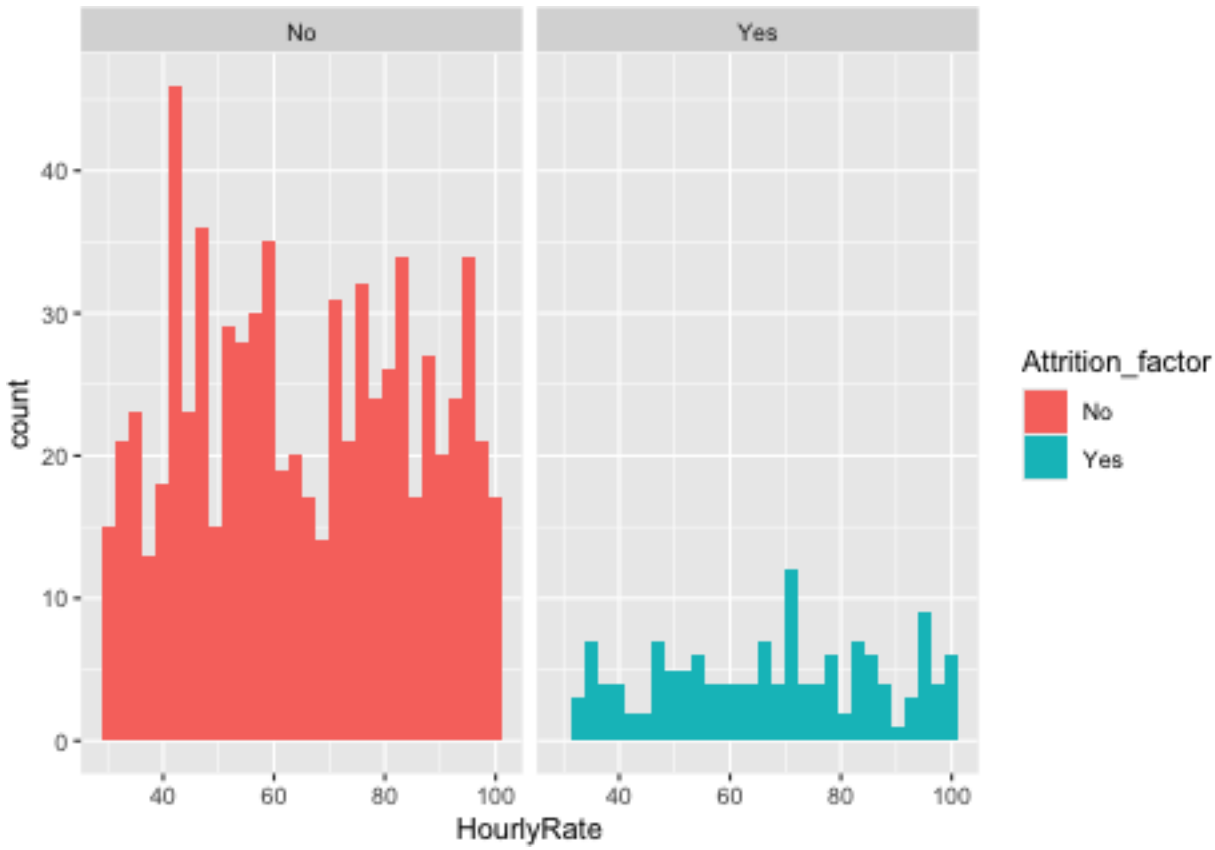
Gender	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Female	301	53	354	15.0%
Male	429	87	516	16.9%

```r
plot_df_7 <- employee_data_clean %>%
  count(Gender_factor, Attrition_factor, name = "n") %>%
  group_by(Gender_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_7, aes(x = Gender_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Gender",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```




```r
## Hourly Rate against Attrition
ggplot(employee_data_clean, aes(x = HourlyRate, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



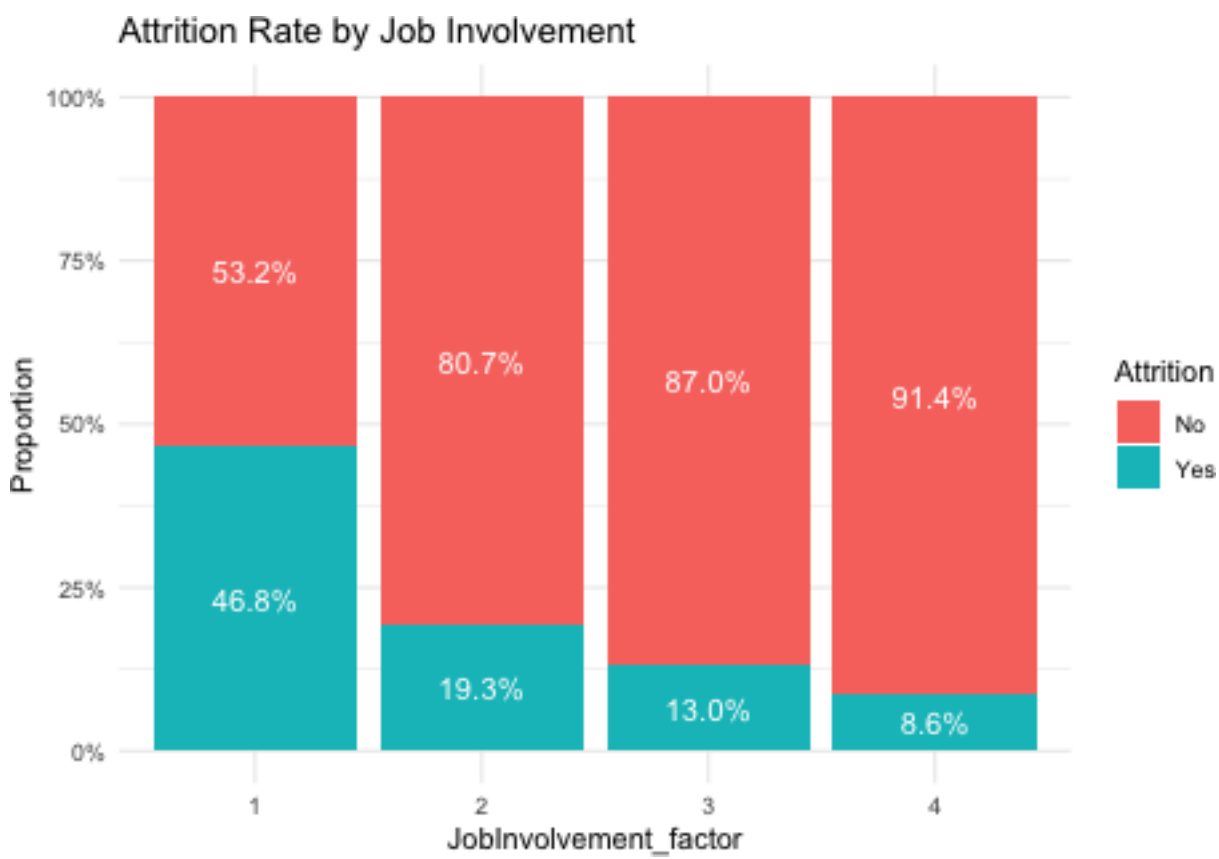

```r
## Job Involvement as related to Attrition

pivot_table <- employee_data_clean %>%
  count(JobInvolvement, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Job Involvement",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Job Involvement")
```

Table 7: Attrition Rate by Job Involvement

Job Involvement	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	25	22	47	46.8%
2	184	44	228	19.3%

Job Involvement	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
3	447	67	514	13.0%
4	74	7	81	8.6%

```r
plot_df_8 <- employee_data_clean %>%
  count(JobInvolvement_factor, Attrition_factor, name = "n") %>%
  group_by(JobInvolvement_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_8, aes(x = JobInvolvement_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Job Involvement",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```

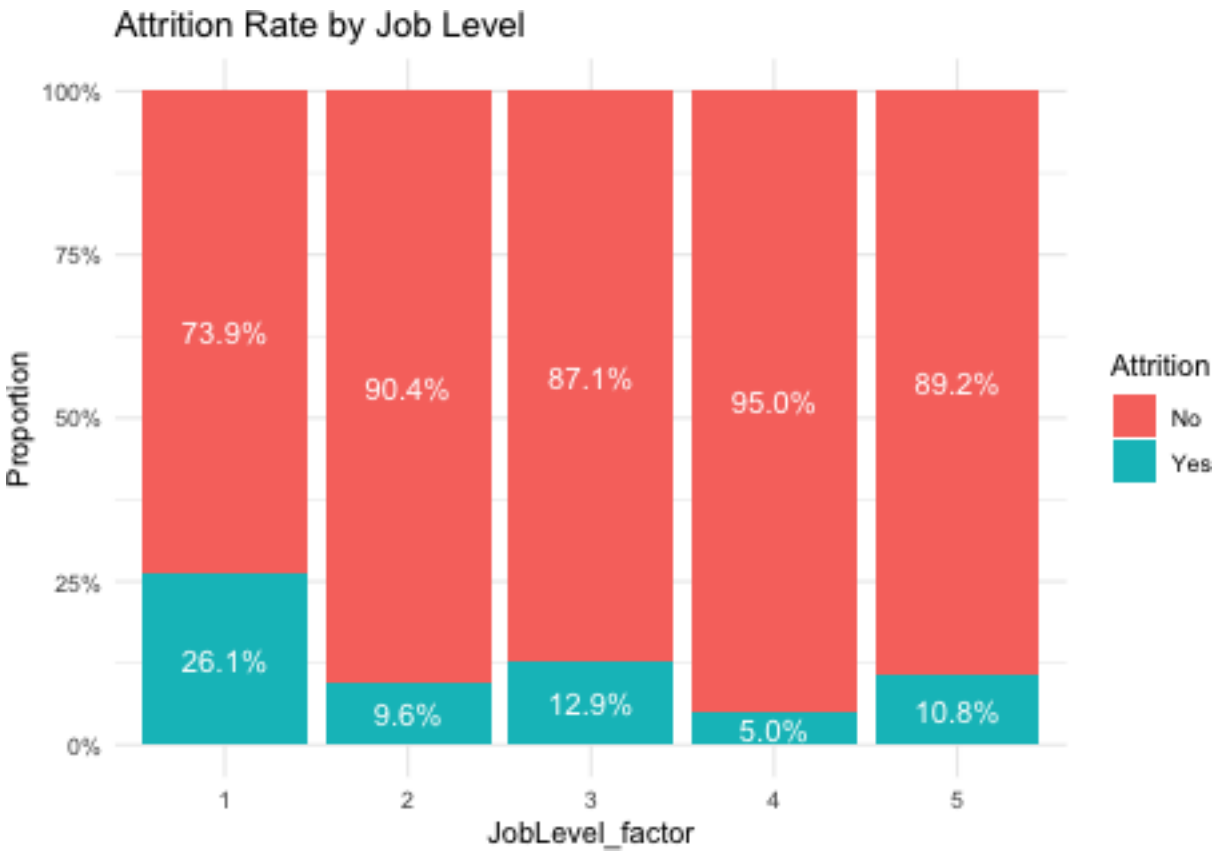


```r
## Job Level related to Attrition

pivot_table <- employee_data_clean %>%
  count(JobLevel, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Job Level",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Job Level")
```

Table 8: Attrition Rate by Job Level

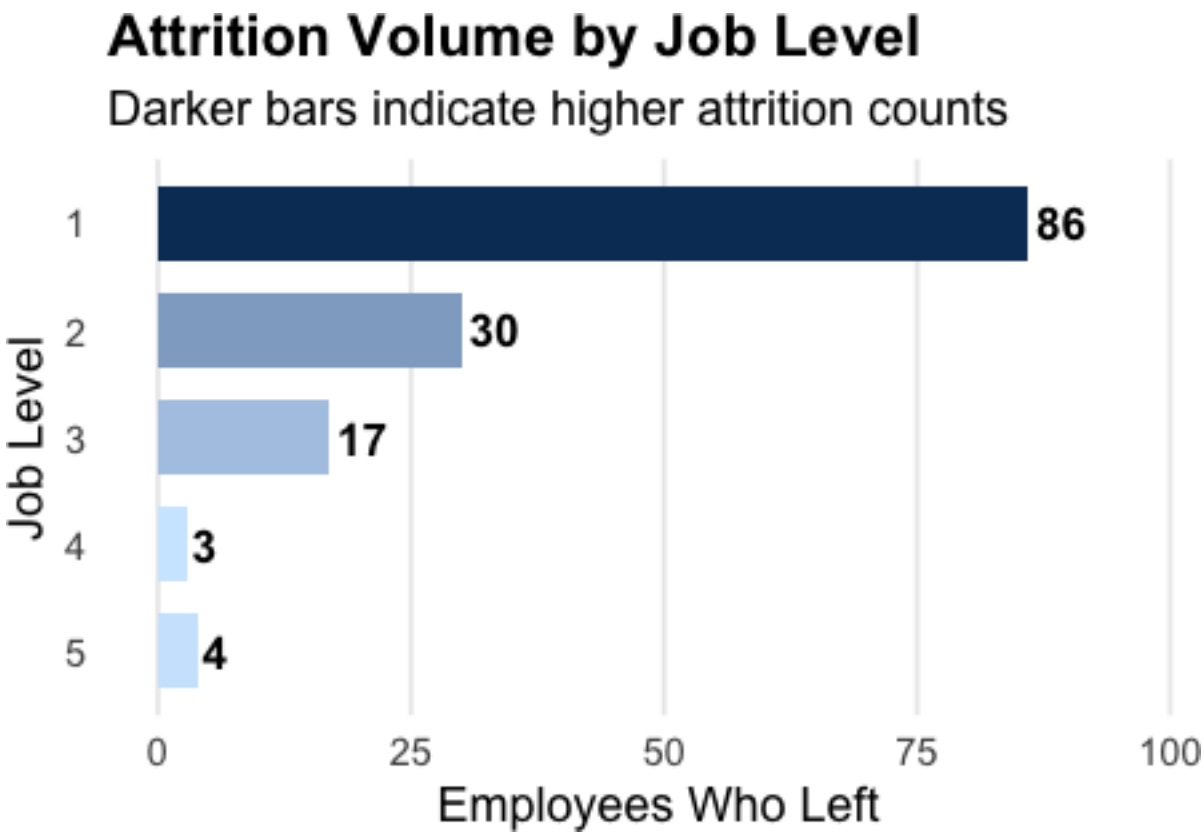
Job Level	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	243	86	329	26.1%
2	282	30	312	9.6%
3	115	17	132	12.9%
4	57	3	60	5.0%
5	33	4	37	10.8%

```r
plot_df_9 <- employee_data_clean %>%
  count(JobLevel_factor, Attrition_factor, name = "n") %>%
  group_by(JobLevel_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_9, aes(x = JobLevel_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Job Level",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```




```r
## Using Volume create a horizontal bar chart of employees who attrit

plot_left <- employee_data_clean %>%
  filter(Attrition_factor == "Yes") %>%
  count(JobLevel, name = "Left_Count") %>%
  mutate(
    JobLevel = factor(JobLevel, levels = 5:1)  # reverse order
  )

ggplot(plot_left, aes(x = JobLevel, y = Left_Count, fill = Left_Count)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = Left_Count),
            hjust = -0.15,
            size = 6,
            fontface = "bold") +
  coord_flip() +
  scale_fill_gradient(low = "#CFE8FF", high = "#0B3A66") +
  labs(
    title = "Attrition Volume by Job Level",
    subtitle = "Darker bars indicate higher attrition counts",
    x = "Job Level",
    y = "Employees Who Left"
  ) +
  theme_minimal(base_size = 18) +
  theme(
    legend.position = "none",
    panel.grid.minor = element_blank(),
```

```r
    panel.grid.major.y = element_blank(),
    plot.title = element_text(face = "bold")
  ) +
  expand_limits(y = max(plot_left$Left_Count) * 1.15)
```

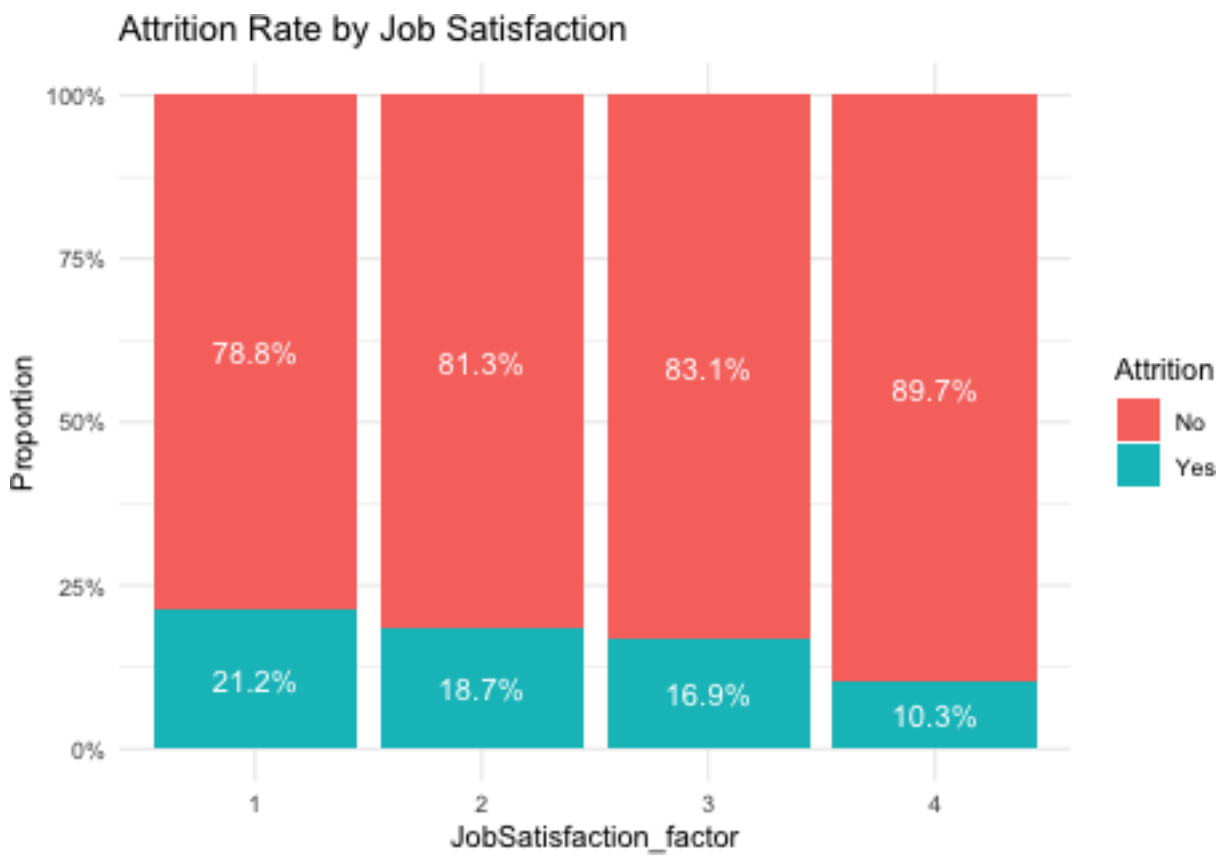



```r
## Job Satisfaction by Attrition

pivot_table <- employee_data_clean %>%
  count(JobSatisfaction, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Job Satisfaction",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Job Satisfaction")
```

Table 9: Attrition Rate by Job Satisfaction

Job Satisfaction	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	141	38	179	21.2%
2	135	31	166	18.7%
3	211	43	254	16.9%
4	243	28	271	10.3%

```r
plot_df_10 <- employee_data_clean %>%
  count(JobSatisfaction_factor, Attrition_factor, name = "n") %>%
  group_by(JobSatisfaction_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_10, aes(x = JobSatisfaction_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Job Satisfaction",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



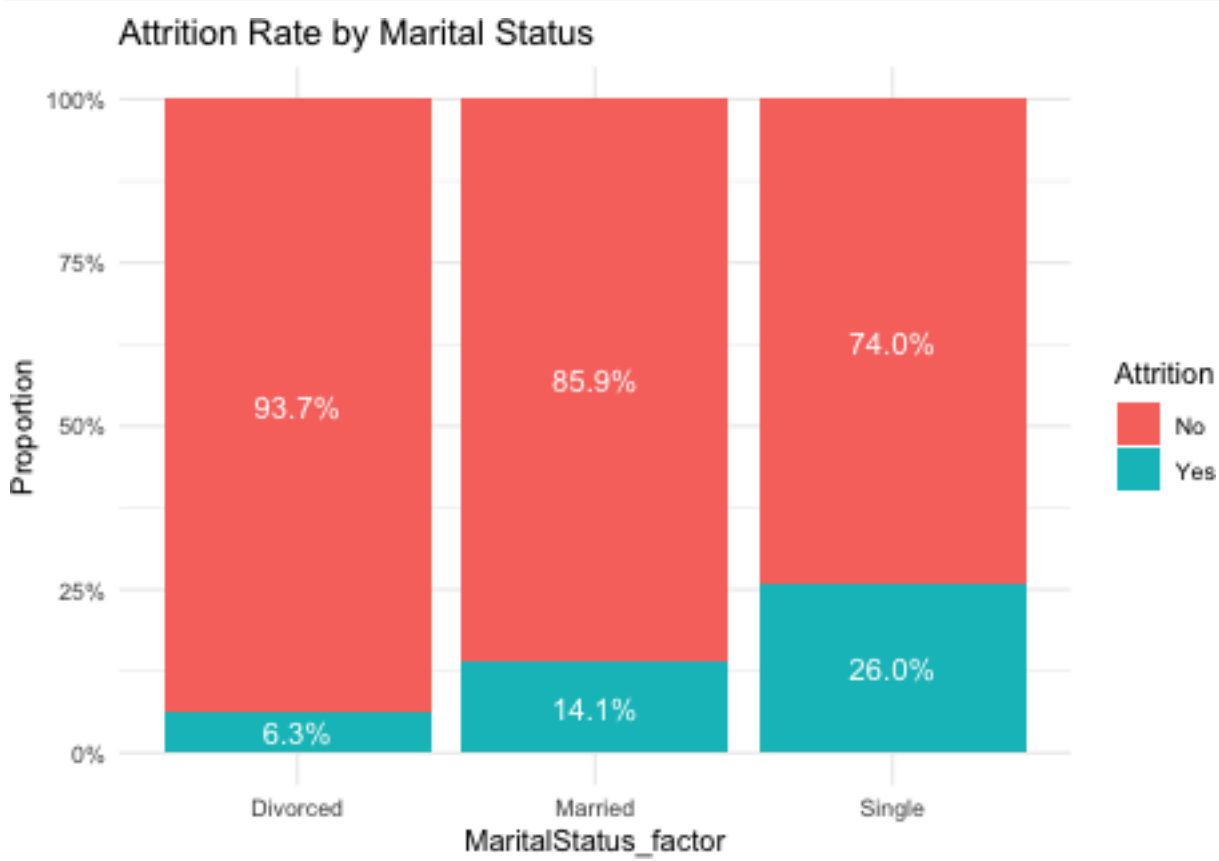

```r
## Marital Status

pivot_table <- employee_data_clean %>%
  count(MaritalStatus, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Marital Status",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Marital Status")
```

Table 10: Attrition Rate by Marital Status

Marital Status	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Divorced	179	12	191	6.3%
Married	352	58	410	14.1%

Marital Status	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
Single	199	70	269	26.0%

```
plot_df_11 <- employee_data_clean %>%
  count(MaritalStatus_factor, Attrition_factor, name = "n") %>%
  group_by(MaritalStatus_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_11, aes(x = MaritalStatus_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Marital Status",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



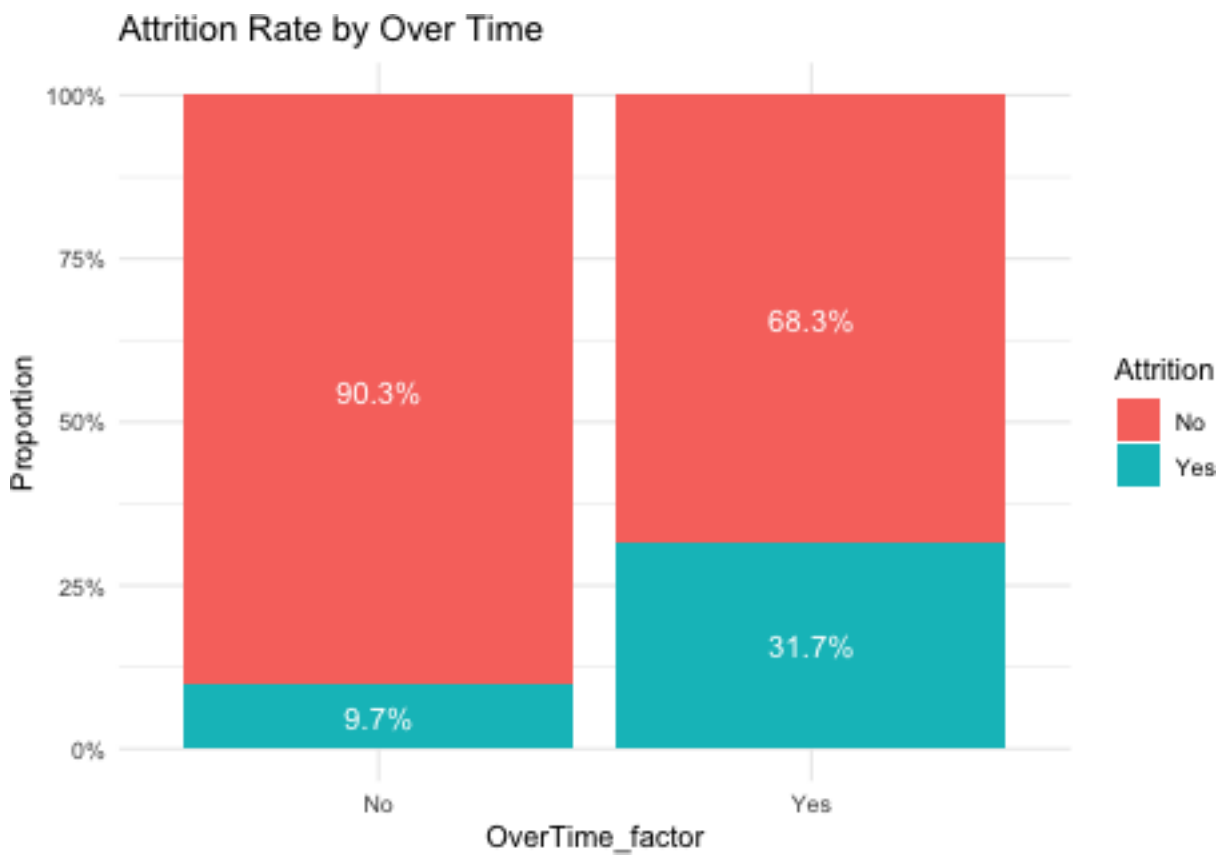

```
## Over time work related to Attrition
```

```r
pivot_table <- employee_data_clean %>%
  count(OverTime, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Over Time",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Over Time")
```

Table 11: Attrition Rate by Over Time

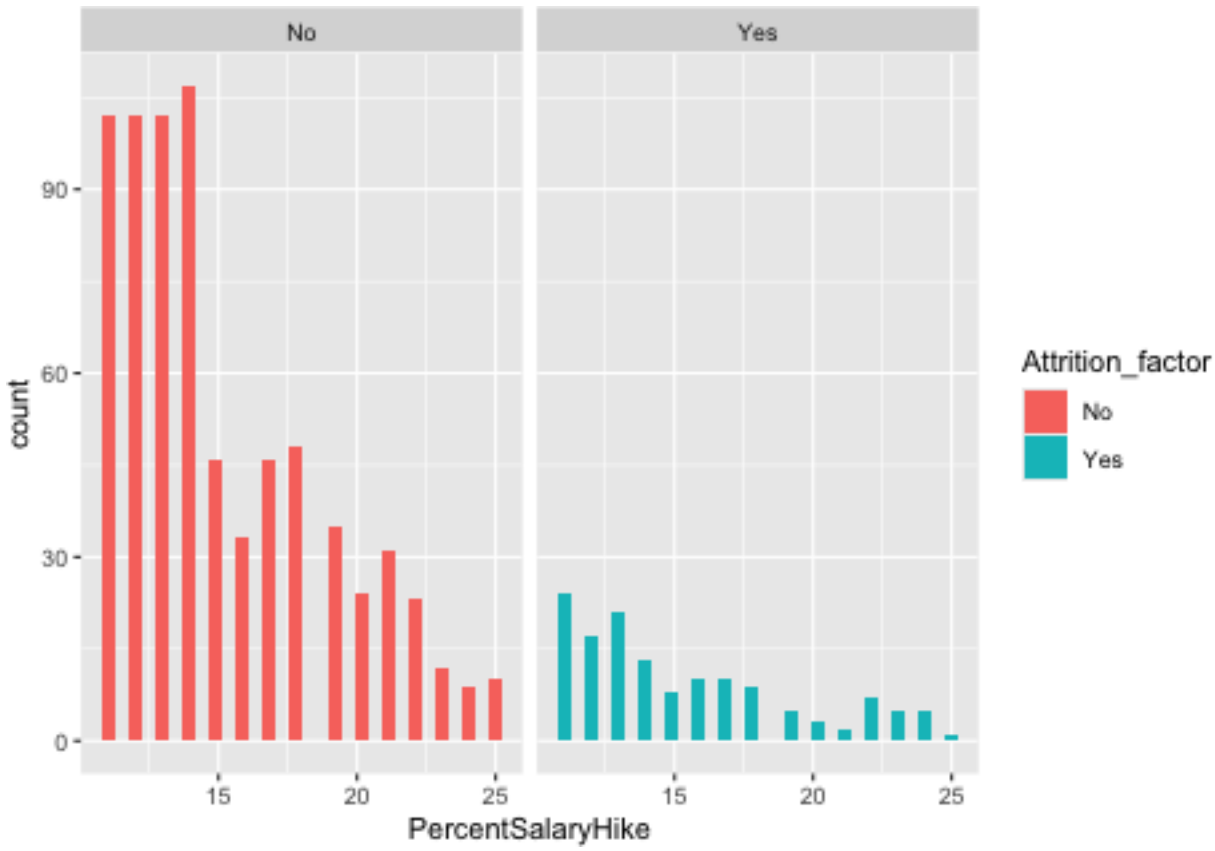
Over Time	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
No	558	60	618	9.7%
Yes	172	80	252	31.7%

```r
plot_df_12 <- employee_data_clean %>%
  count(OverTime_factor, Attrition_factor, name = "n") %>%
  group_by(OverTime_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_12, aes(x = OverTime_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Over Time",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```




```r
## Percentage Increase related to Attrition

ggplot(employee_data_clean, aes(x = PercentSalaryHike, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
## Performance Rating related to Attrition. This is an interesting self-reporting statistic as no emplo

pivot_table <- employee_data_clean %>%
  count(PerformanceRating, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Performance Rating",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Performance Rating")
```

Table 12: Attrition Rate by Performance Rating

Performance Rating	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
3	621	117	738	15.9%
4	109	23	132	17.4%

```r
plot_df_13 <- employee_data_clean %>%
  count(PerformanceRating_factor, Attrition_factor, name = "n") %>%
  group_by(PerformanceRating_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_13, aes(x = PerformanceRating_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Performance Rating",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```

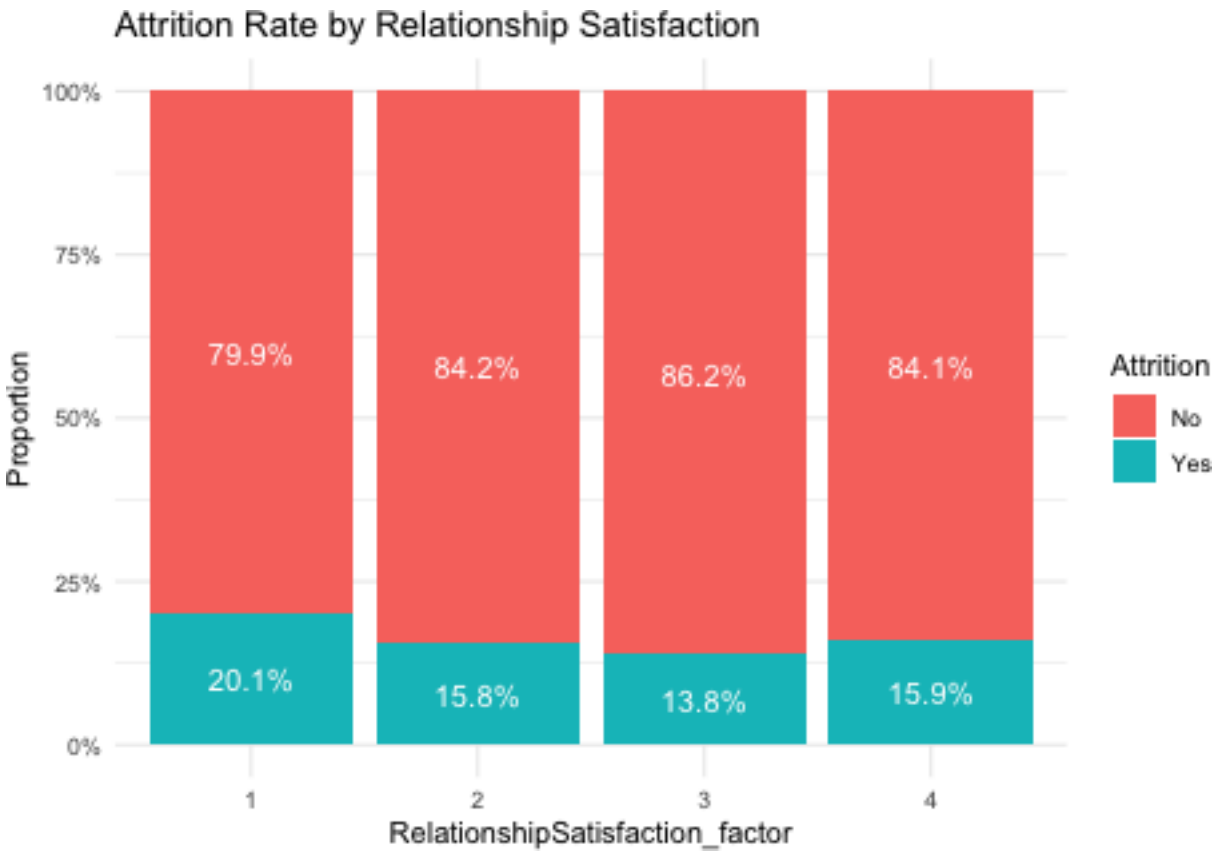
```r
## Relationship Satisfaction related to Attrition

pivot_table <- employee_data_clean %>%
  count(RelationshipSatisfaction, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
```

```r
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Relationship Satisfaction",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Relationship Satisfaction")
```

Table 13: Attrition Rate by Relationship Satisfaction

Relationship Satisfaction	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	139	35	174	20.1%
2	144	27	171	15.8%
3	225	36	261	13.8%
4	222	42	264	15.9%

```r
plot_df_14 <- employee_data_clean %>%
  count(RelationshipSatisfaction_factor, Attrition_factor, name = "n") %>%
  group_by(RelationshipSatisfaction_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_14, aes(x = RelationshipSatisfaction_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Relationship Satisfaction",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



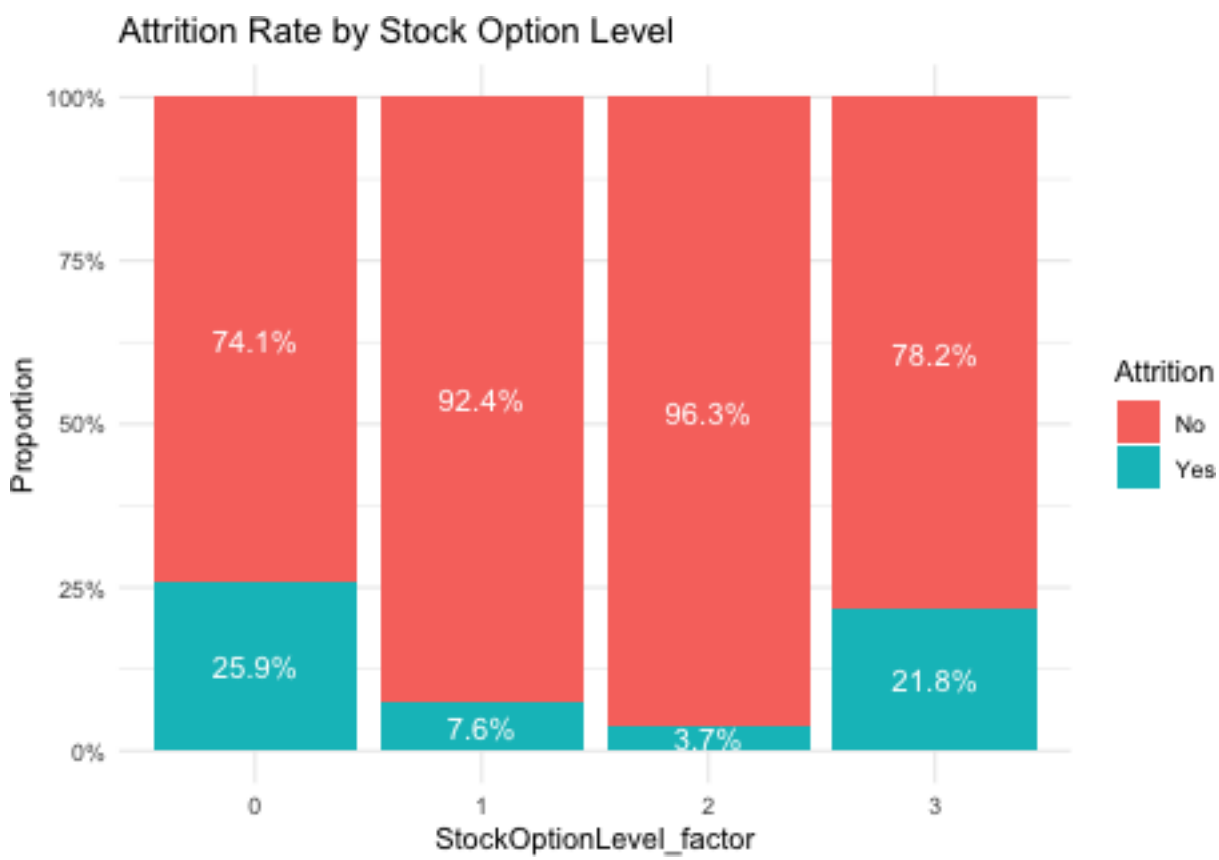

```r
## Stock Option Level

pivot_table <- employee_data_clean %>%
  count(StockOptionLevel, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Stock Option Level",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
      caption = "Attrition Rate by Stock Option Level")
```

Table 14: Attrition Rate by Stock Option Level

Stock Option Level	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
0	281	98	379	25.9%
1	328	27	355	7.6%

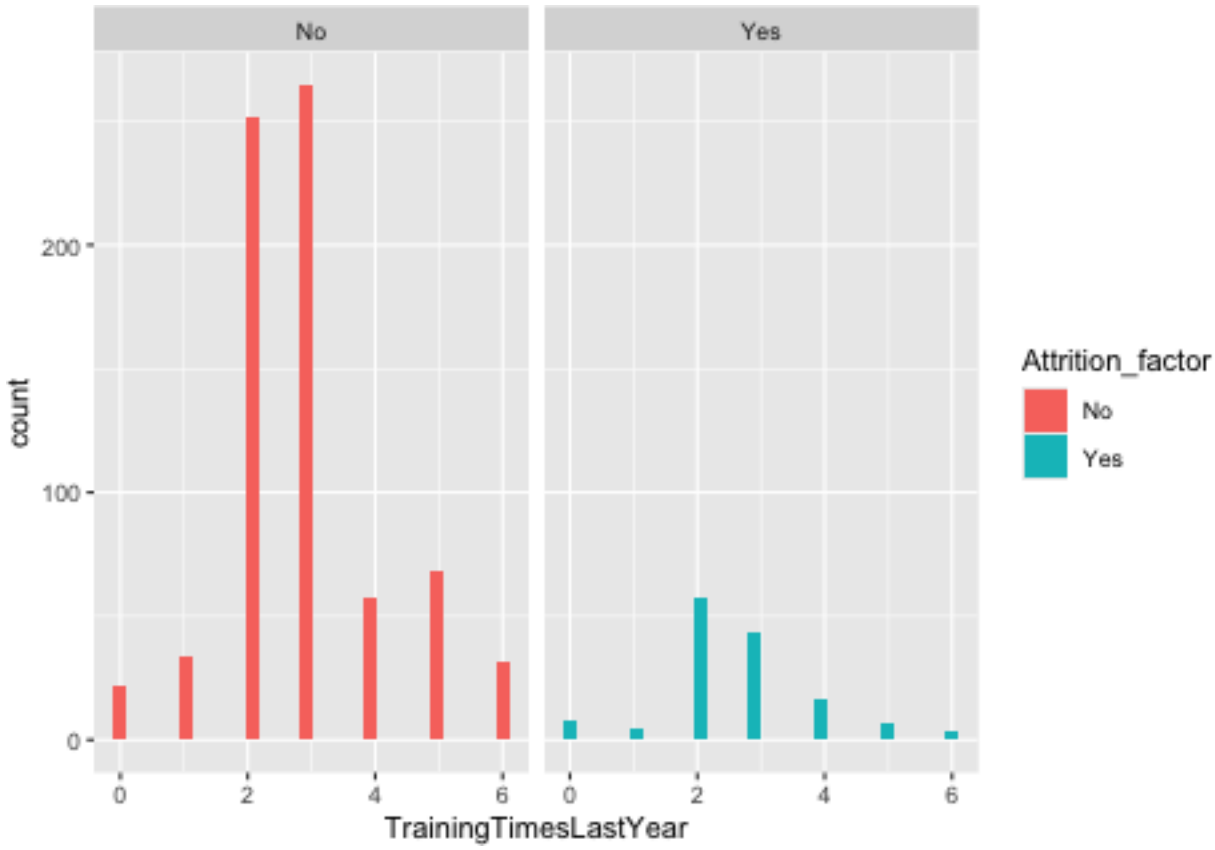
Stock Option Level	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
2	78	3	81	3.7%
3	43	12	55	21.8%

```r
plot_df_15 <- employee_data_clean %>%
  count(StockOptionLevel_factor, Attrition_factor, name = "n") %>%
  group_by(StockOptionLevel_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_15, aes(x = StockOptionLevel_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Stock Option Level",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



```r
## Training Times Last Year

ggplot(employee_data_clean, aes(x = TrainingTimesLastYear, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



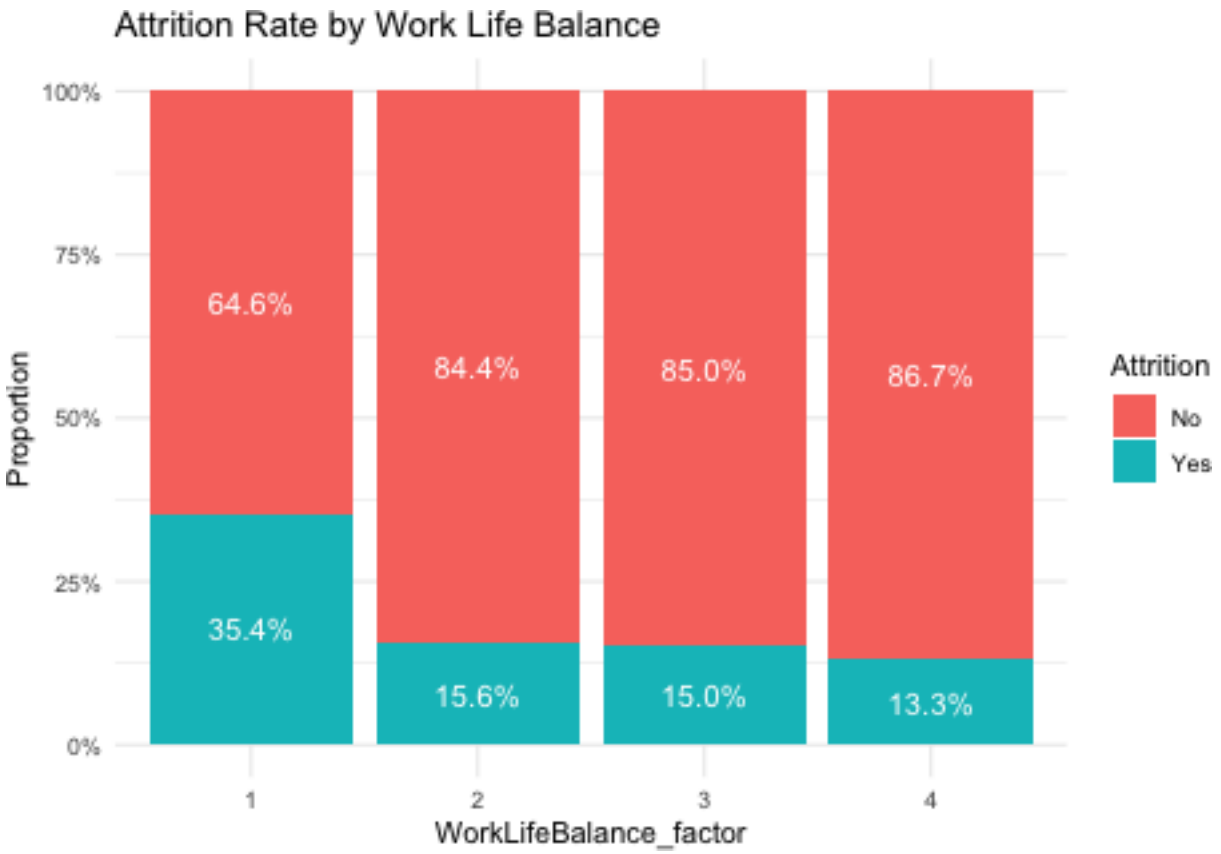

```r
##Work Life Balance related to Attrition

pivot_table <- employee_data_clean %>%
  count(WorkLifeBalance, Attrition_factor) %>%
  pivot_wider(
    names_from = Attrition_factor,
    values_from = n
  ) %>%
  mutate(
    Total = No + Yes,
    Attrition_Rate = percent(Yes / Total, accuracy = 0.1)
  )

kable(pivot_table,
      col.names = c("Work Life Balance",
                    "Stayed (No)",
                    "Attrited (Yes)",
                    "Total Employees",
                    "Attrition Rate"),
```

```
      caption = "Attrition Rate by Work Life Balance")
```

Table 15: Attrition Rate by Work Life Balance

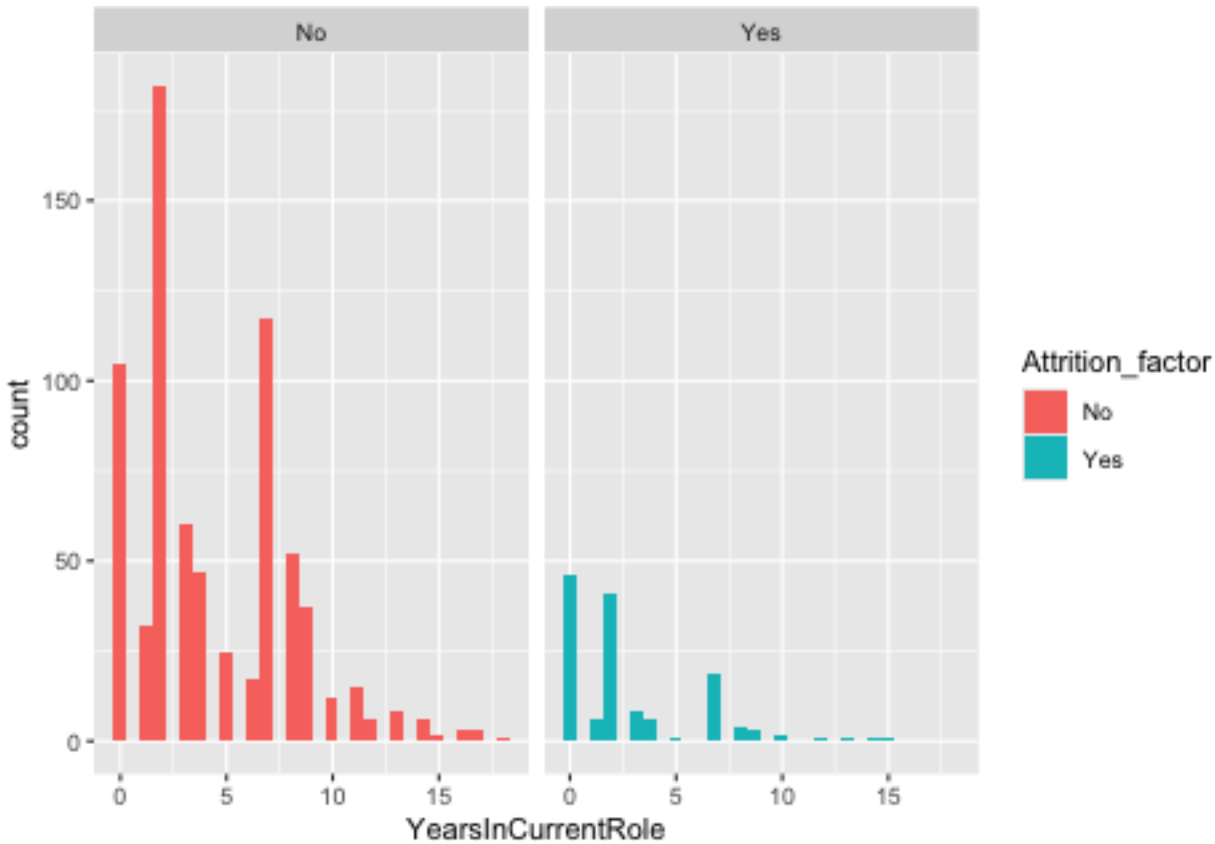
Work Life Balance	Stayed (No)	Attrited (Yes)	Total Employees	Attrition Rate
1	31	17	48	35.4%
2	162	30	192	15.6%
3	452	80	532	15.0%
4	85	13	98	13.3%

```
plot_df_16 <- employee_data_clean %>%
  count(WorkLifeBalance_factor, Attrition_factor, name = "n") %>%
  group_by(WorkLifeBalance_factor) %>%
  mutate(
    prop  = n / sum(n),
    label = percent(prop, accuracy = 0.1)
  ) %>%
  ungroup()

ggplot(plot_df_16, aes(x = WorkLifeBalance_factor, y = prop, fill = Attrition_factor)) +
  geom_col() +
  geom_text(aes(label = label),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 4) +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Attrition Rate by Work Life Balance",
       y = "Proportion",
       fill = "Attrition") +
  theme_minimal()
```



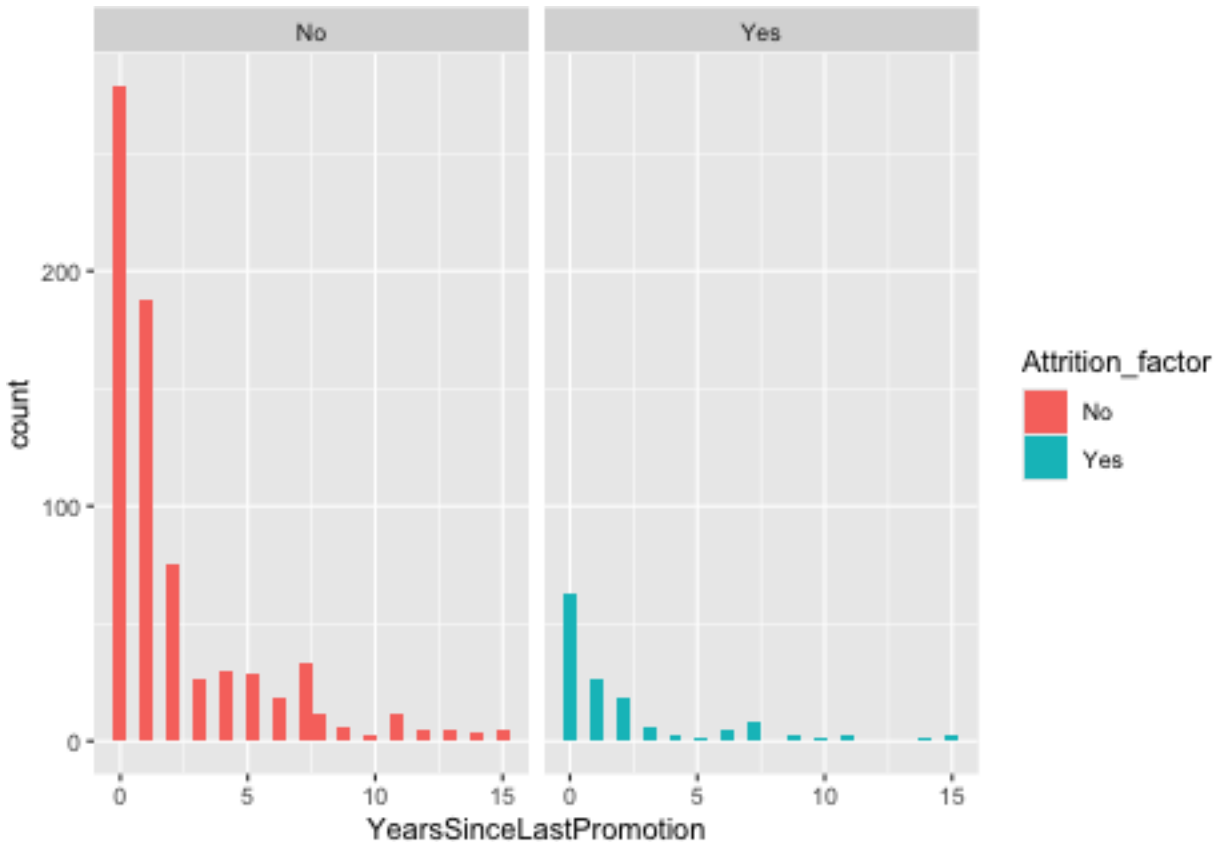

```r
## Years in Current Role related to Attrition

ggplot(employee_data_clean, aes(x = YearsInCurrentRole, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



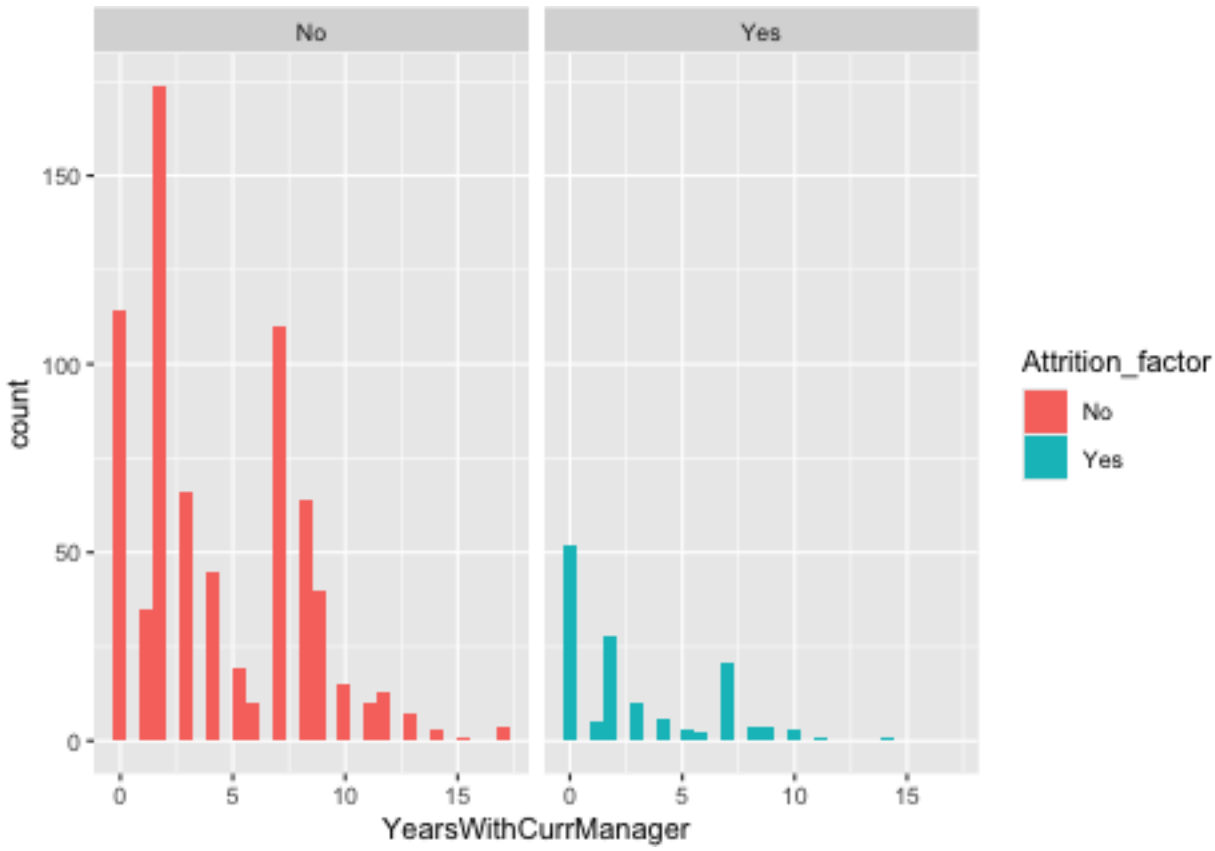

```r
## Years since last promotion

ggplot(employee_data_clean, aes(x = YearsSinceLastPromotion, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
## Years with current manager

ggplot(employee_data_clean, aes(x = YearsWithCurrManager, fill = Attrition_factor)) +
  geom_histogram() +
  facet_wrap(vars(Attrition_factor))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
## These visualizations provide a path forward but do not deal with issues around the scale of the data

## Rank Variables by number of employees (volume) and probability in descending order

attrition_counts <- employee_data_clean %>%
  filter(Attrition == "Yes") %>%
  select(where(is.factor)) %>%   # only factor variables
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Value") %>%
  group_by(Variable, Value) %>%
  summarise(employees_lost = n(), .groups = "drop") %>%
  arrange(desc(employees_lost))

# View the top attrition drivers
view(attrition_counts)

attrition_probability <- employee_data_clean %>%
  select(Attrition, where(is.factor)) %>%   # keep Attrition
  pivot_longer(cols = -Attrition,
               names_to = "Variable",
               values_to = "Value") %>%
  group_by(Variable, Value) %>%
  summarise(
    total_employees = n(),
    employees_lost = sum(Attrition == "Yes"),
    attrition_rate = employees_lost / total_employees,
```

```r
    .groups = "drop"
  ) %>%
  arrange(desc(attrition_rate))

view(attrition_probability)
```

Reviewing the impact of career stage on the top 3 attrition factors

```r
attrition_profile_plot = employee_data_clean %>%
  mutate(
    CareerStage = case_when(
      TotalWorkingYears <= 5 ~ "Early Career",
      TotalWorkingYears <= 10 ~ "Developing",
      TotalWorkingYears <= 20 ~ "Established",
      TRUE ~ "Senior"
    ),
    StockOptionGroup = ifelse(StockOptionLevel == 0, "No Stock Options", "Has Stock Options")
  ) %>%
  group_by(CareerStage, OverTime, StockOptionGroup) %>%
  summarise(
    AttritionRate = mean(Attrition_factor == "Yes"),
    Count = n(),
    .groups = "drop"
  ) %>%
  ggplot(aes(x = CareerStage, y = AttritionRate, fill = OverTime)) +
  geom_col(position = "dodge") +
  facet_wrap(~StockOptionGroup) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Attrition Risk by Employee Profile",
    subtitle = "Career Stage, Overtime, and Stock Incentives",
    x = "Career Stage",
    y = "Attrition Rate",
    fill = "Overtime"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

attrition_profile_plot
```

Data Wrangling for k-NN model

```
## I kept the ordinal variables numeric and scaled because k-NN works off distance. If I dummy-encoded

## First dataset with full values to test the quality of the datasets
  kNN_employee_data <- employee_data_clean %>%
  mutate(
    Attrition_factor = factor(Attrition_factor, levels = c("No", "Yes")),
    OverTime_Yes = ifelse(OverTime == "Yes", 1, 0),
    MaritalStatus_Single = ifelse(MaritalStatus == "Single", 1, 0),
    Travel_Frequently = ifelse(BusinessTravel == "Travel_Frequently", 1, 0)
  ) %>%
  select(
    Attrition_factor,
    Age,
    TotalWorkingYears,
    MonthlyIncome,
    StockOptionLevel,
    JobLevel,
    JobSatisfaction,
    JobInvolvement,
    WorkLifeBalance,
    OverTime_Yes,
    MaritalStatus_Single,
    Travel_Frequently)

names(kNN_employee_data)
```

```
##  [1] "Attrition_factor"     "Age"                  "TotalWorkingYears"
##  [4] "MonthlyIncome"        "StockOptionLevel"     "JobLevel"
##  [7] "JobSatisfaction"      "JobInvolvement"       "WorkLifeBalance"
## [10] "OverTime_Yes"         "MaritalStatus_Single" "Travel_Frequently"
```

```
 kNN_small_data <- employee_data_clean %>%
  mutate(
    Attrition_factor = factor(Attrition_factor, levels = c("No", "Yes")),
    OverTime_Yes = ifelse(OverTime == "Yes", 1, 0),
    MaritalStatus_Single = ifelse(MaritalStatus == "Single", 1, 0),
    Travel_Frequently = ifelse(BusinessTravel == "Travel_Frequently", 1, 0)
  ) %>%
  select(
  Attrition_factor,
  StockOptionLevel,
  JobLevel,
  OverTime_Yes,
  MonthlyIncome,
  Age
)

 view(kNN_small_data)

## Scale data
kNN_data <- kNN_employee_data %>%
  mutate(across(-Attrition_factor, scale))

kNN_data2 <- kNN_small_data %>%
  mutate(across(-Attrition_factor,scale))

## Create test and train datasets
set.seed(123)
trainIndices <- createDataPartition(
  employee_data_clean$Attrition_factor,
  p = 0.70,
  list = FALSE
)

trainData_knn <- kNN_data[trainIndices, ]
testData_knn  <- kNN_data[-trainIndices, ]

head(trainData_knn)
```

```
##    Attrition_factor         Age TotalWorkingYears MonthlyIncome
## 5                No -1.43724030        -0.6724910   -0.57208314
## 6                No -1.10114163        -0.2732186    0.52259559
## 8                No  0.01918724        -0.4063094    0.06606257
## 10               No -0.31691142        -0.4063094   -0.28868032
## 11               No  0.69138457         1.3238710    2.82788068
## 12               No -0.98910875        -1.0717634   -0.80132840
##    StockOptionLevel    JobLevel JobSatisfaction JobInvolvement WorkLifeBalance
## 5        -0.9138971 -0.95318680       1.1581741       0.393688       0.3067386
## 6         1.4177465  0.88148691      -1.5335752       0.393688      -1.0978013
## 8         2.5835682 -0.03584995       1.1581741      -1.027509       0.3067386
## 10        0.2519247 -0.03584995       0.2609244      -1.027509      -1.0978013
```

```
## 11       -0.9138971  2.71616062       0.2609244      -1.027509       0.3067386
## 12        0.2519247 -0.95318680       1.1581741       0.393688       0.3067386
##    OverTime_Yes MaritalStatus_Single Travel_Frequently
## 5     1.5651077            1.4938650         2.1215912
## 6    -0.6381993           -0.6686351         2.1215912
## 8     1.5651077           -0.6686351        -0.4708026
## 10   -0.6381993           -0.6686351         2.1215912
## 11   -0.6381993           -0.6686351        -0.4708026
## 12   -0.6381993           -0.6686351        -0.4708026
```

```r
head(testData_knn)
```

```
##   Attrition_factor        Age TotalWorkingYears MonthlyIncome StockOptionLevel
## 1               No -0.5409772        -0.4063094    -0.4322305        0.2519247
## 2               No  0.3552859         1.3238710     2.8787757       -0.9138971
## 3               No -0.2048785        -0.1401278     0.6463532       -0.9138971
## 4               No -0.5409772         0.3922354     0.8769035        1.4177465
## 7               No  0.4673188        -0.5394002    -0.9272610       -0.9138971
## 9               No -0.3169114        -1.3379450    -0.9070335        0.2519247
##      JobLevel JobSatisfaction JobInvolvement WorkLifeBalance OverTime_Yes
## 1 -0.03584995       1.1581741       0.393688      -1.0978013   -0.6381993
## 2  2.71616062       0.2609244      -1.027509       1.7112786   -0.6381993
## 3  0.88148691       1.1581741       0.393688       0.3067386   -0.6381993
## 4  0.88148691       1.1581741       0.393688       0.3067386   -0.6381993
## 7 -0.95318680       0.2609244       1.814885      -1.0978013    1.5651077
## 9 -0.95318680       0.2609244       0.393688       0.3067386    1.5651077
##   MaritalStatus_Single Travel_Frequently
## 1           -0.6686351        -0.4708026
## 2            1.4938650        -0.4708026
## 3            1.4938650         2.1215912
## 4           -0.6686351        -0.4708026
## 7           -0.6686351        -0.4708026
## 9           -0.6686351        -0.4708026
```

```r
names(trainData_knn)
```

```
##  [1] "Attrition_factor"     "Age"                  "TotalWorkingYears"
##  [4] "MonthlyIncome"        "StockOptionLevel"     "JobLevel"
##  [7] "JobSatisfaction"      "JobInvolvement"       "WorkLifeBalance"
## [10] "OverTime_Yes"         "MaritalStatus_Single" "Travel_Frequently"
```

```r
trainData_knn2 <- kNN_data2[trainIndices, ]
testData_knn2  <- kNN_data2[-trainIndices, ]

head(trainData_knn2)
```

```
##    Attrition_factor StockOptionLevel    JobLevel OverTime_Yes MonthlyIncome
## 5                No       -0.9138971 -0.95318680    1.5651077   -0.57208314
## 6                No        1.4177465  0.88148691   -0.6381993    0.52259559
## 8                No        2.5835682 -0.03584995    1.5651077    0.06606257
## 10               No        0.2519247 -0.03584995   -0.6381993   -0.28868032
## 11               No       -0.9138971  2.71616062   -0.6381993    2.82788068
## 12               No        0.2519247 -0.95318680   -0.6381993   -0.80132840
##            Age
## 5  -1.43724030
## 6  -1.10114163
```

```
## 8   0.01918724
## 10 -0.31691142
## 11  0.69138457
## 12 -0.98910875
```

```r
head(testData_knn2)
```

```
##   Attrition_factor StockOptionLevel    JobLevel OverTime_Yes MonthlyIncome
## 1               No        0.2519247 -0.03584995   -0.6381993    -0.4322305
## 2               No       -0.9138971  2.71616062   -0.6381993     2.8787757
## 3               No       -0.9138971  0.88148691   -0.6381993     0.6463532
## 4               No        1.4177465  0.88148691   -0.6381993     0.8769035
## 7               No       -0.9138971 -0.95318680    1.5651077    -0.9272610
## 9               No        0.2519247 -0.95318680    1.5651077    -0.9070335
##          Age
## 1 -0.5409772
## 2  0.3552859
## 3 -0.2048785
## 4 -0.5409772
## 7  0.4673188
## 9 -0.3169114
```

```r
names(trainData_knn2)
```

```
## [1] "Attrition_factor" "StockOptionLevel" "JobLevel"         "OverTime_Yes"
## [5] "MonthlyIncome"    "Age"
```

```r
knn_k5 <- train(
  Attrition_factor ~ .,
  data = trainData_knn,
  method = "knn",
  preProcess = c("center", "scale"),
  tuneGrid = data.frame(k = 5)
)

knn_k5
```

```
## k-Nearest Neighbors
##
## 609 samples
##  11 predictor
##   2 classes: 'No', 'Yes'
##
## Pre-processing: centered (11), scaled (11)
## Resampling: Bootstrapped (25 reps)
## Summary of sample sizes: 609, 609, 609, 609, 609, 609, ...
## Resampling results:
##
##   Accuracy   Kappa
##   0.8181444  0.2292031
##
## Tuning parameter 'k' was held constant at a value of 5
```

```r
knn_k5b <- train(
  Attrition_factor ~ .,
  data = trainData_knn2,
  method = "knn",
```

```r
  preProcess = c("center", "scale"),
  tuneGrid = data.frame(k = 5)
)

knn_k5b
```

```
## k-Nearest Neighbors
##
## 609 samples
##   5 predictor
##   2 classes: 'No', 'Yes'
##
## Pre-processing: centered (5), scaled (5)
## Resampling: Bootstrapped (25 reps)
## Summary of sample sizes: 609, 609, 609, 609, 609, 609, ...
## Resampling results:
##
##   Accuracy   Kappa
##   0.8266805  0.2938367
##
## Tuning parameter 'k' was held constant at a value of 5
```

```r
pred_class <- predict(knn_k5, newdata = testData_knn)
confusionMatrix(pred_class, testData_knn$Attrition_factor, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  211  35
##        Yes   8   7
##
##                Accuracy : 0.8352
##                  95% CI : (0.7846, 0.8781)
##     No Information Rate : 0.8391
##     P-Value [Acc > NIR] : 0.6066
##
##                   Kappa : 0.1758
##
##  Mcnemar's Test P-Value : 7.341e-05
##
##             Sensitivity : 0.16667
##             Specificity : 0.96347
##          Pos Pred Value : 0.46667
##          Neg Pred Value : 0.85772
##              Prevalence : 0.16092
##          Detection Rate : 0.02682
##    Detection Prevalence : 0.05747
##       Balanced Accuracy : 0.56507
##
##        'Positive' Class : Yes
##
```

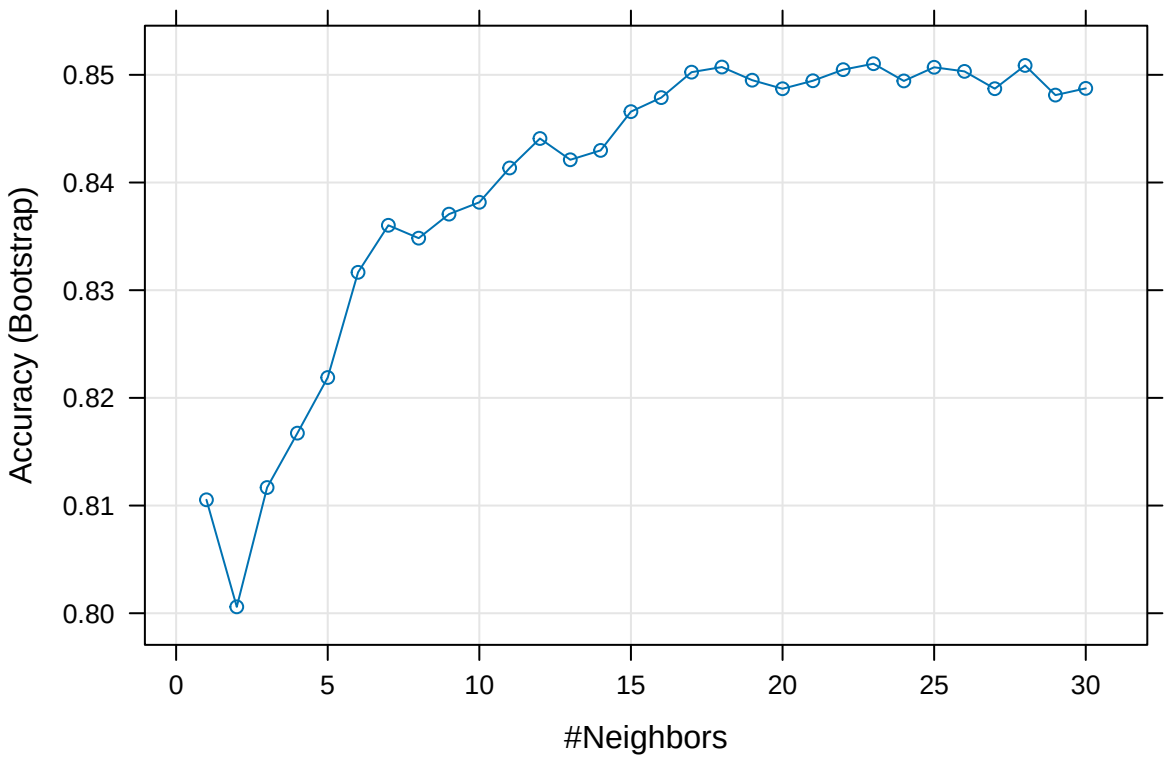
```r
pred_class <- predict(knn_k5b, newdata = testData_knn2)
confusionMatrix(pred_class, testData_knn2$Attrition_factor, positive = "Yes")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  206  31
##        Yes  13  11
##
##                Accuracy : 0.8314
##                  95% CI : (0.7804, 0.8748)
##     No Information Rate : 0.8391
##     P-Value [Acc > NIR] : 0.66882
##
##                   Kappa : 0.245
##
##  Mcnemar's Test P-Value : 0.01038
##
##             Sensitivity : 0.26190
##             Specificity : 0.94064
##          Pos Pred Value : 0.45833
##          Neg Pred Value : 0.86920
##              Prevalence : 0.16092
##          Detection Rate : 0.04215
##    Detection Prevalence : 0.09195
##       Balanced Accuracy : 0.60127
##
##        'Positive' Class : Yes
##
```

```r
knn_tuned <- train(
  Attrition_factor ~ .,
  data = trainData_knn2,
  method = "knn",
  preProcess = c("center", "scale"),
  tuneGrid = data.frame(k = 1:30)
)

knn_tuned$bestTune
```

```
##     k
## 23 23
```

```r
plot(knn_tuned)
```




```r
pred_class <- predict(knn_tuned, newdata = testData_knn2)
confusionMatrix(pred_class, testData_knn2$Attrition_factor, positive = "Yes")
```

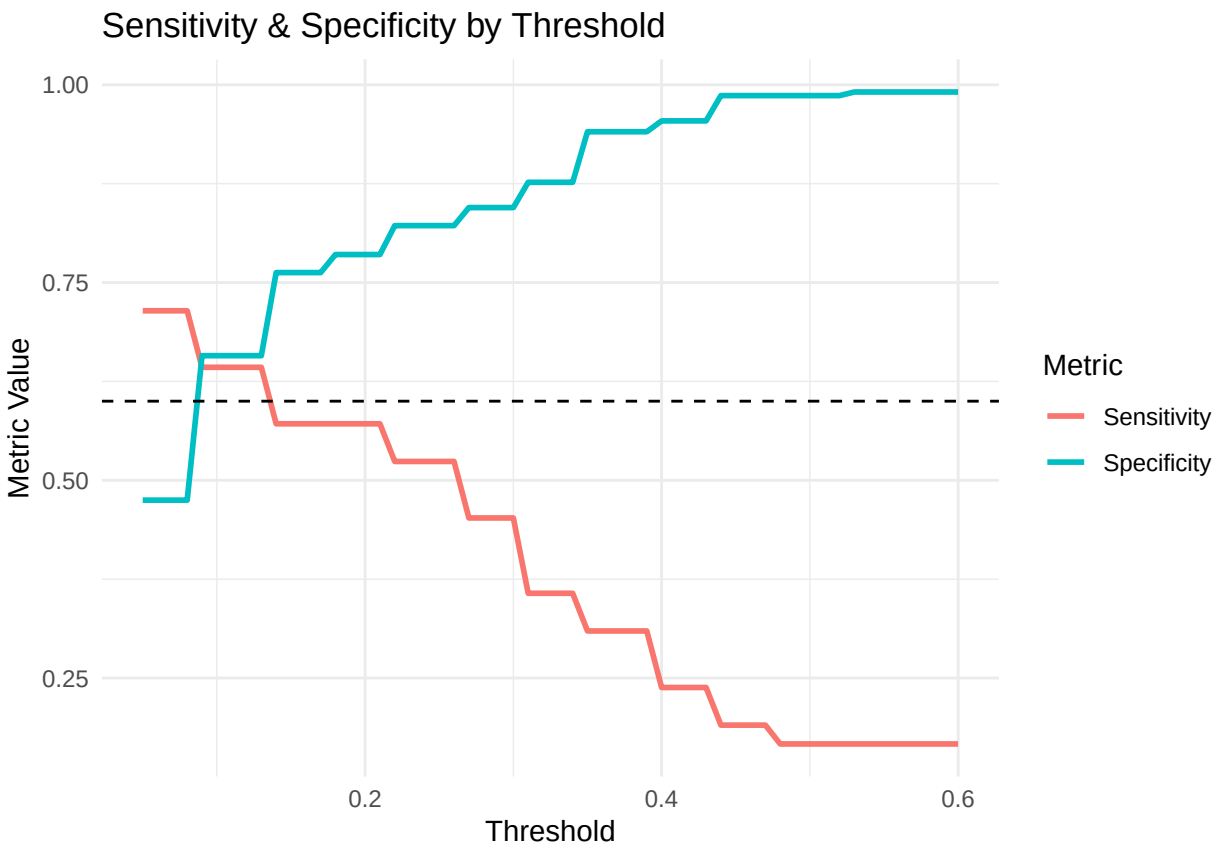
```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  216  35
##        Yes   3   7
##
##                Accuracy : 0.8544
##                  95% CI : (0.8057, 0.8949)
##     No Information Rate : 0.8391
##     P-Value [Acc > NIR] : 0.282
##
##                   Kappa : 0.221
##
##  Mcnemar's Test P-Value : 4.934e-07
##
##             Sensitivity : 0.16667
##             Specificity : 0.98630
##          Pos Pred Value : 0.70000
##          Neg Pred Value : 0.86056
##              Prevalence : 0.16092
##          Detection Rate : 0.02682
##    Detection Prevalence : 0.03831
##       Balanced Accuracy : 0.57648
##
##        'Positive' Class : Yes
##
```

```r
pred_prob <- predict(knn_tuned, newdata = testData_knn2, type = "prob")[, "Yes"]
thresholds <- seq(0.05, 0.60, by = 0.01)

# Based on the initial results of the full and smaller knn datasets, I have chosen to use the smaller o

knn_results <- data.frame(threshold = thresholds, Sensitivity = NA, Specificity = NA, Accuracy = NA, TP

for (i in seq_along(thresholds)) {
  t <- thresholds[i]
  preds <- factor(ifelse(pred_prob >= t, "Yes", "No"), levels = c("No","Yes"))
  cm <- confusionMatrix(preds, testData_knn2$Attrition_factor, positive = "Yes")
  tab <- cm$table
  knn_results$Sensitivity[i] <- unname(cm$byClass["Sensitivity"])
  knn_results$Specificity[i] <- unname(cm$byClass["Specificity"])
  knn_results$Accuracy[i]    <- unname(cm$overall["Accuracy"])
  knn_results$TN[i] <- tab["No","No"];  knn_results$TP[i] <- tab["Yes","Yes"]
  knn_results$FP[i] <- tab["Yes","No"]; knn_results$FN[i] <- tab["No","Yes"]
}

knn_results
```

```
##    threshold Sensitivity Specificity  Accuracy TP  FP  TN FN
## 1       0.05   0.7142857   0.4748858 0.5134100 30 115 104 12
## 2       0.06   0.7142857   0.4748858 0.5134100 30 115 104 12
## 3       0.07   0.7142857   0.4748858 0.5134100 30 115 104 12
## 4       0.08   0.7142857   0.4748858 0.5134100 30 115 104 12
## 5       0.09   0.6428571   0.6575342 0.6551724 27  75 144 15
## 6       0.10   0.6428571   0.6575342 0.6551724 27  75 144 15
## 7       0.11   0.6428571   0.6575342 0.6551724 27  75 144 15
## 8       0.12   0.6428571   0.6575342 0.6551724 27  75 144 15
## 9       0.13   0.6428571   0.6575342 0.6551724 27  75 144 15
## 10      0.14   0.5714286   0.7625571 0.7318008 24  52 167 18
## 11      0.15   0.5714286   0.7625571 0.7318008 24  52 167 18
## 12      0.16   0.5714286   0.7625571 0.7318008 24  52 167 18
## 13      0.17   0.5714286   0.7625571 0.7318008 24  52 167 18
## 14      0.18   0.5714286   0.7853881 0.7509579 24  47 172 18
## 15      0.19   0.5714286   0.7853881 0.7509579 24  47 172 18
## 16      0.20   0.5714286   0.7853881 0.7509579 24  47 172 18
## 17      0.21   0.5714286   0.7853881 0.7509579 24  47 172 18
## 18      0.22   0.5238095   0.8219178 0.7739464 22  39 180 20
## 19      0.23   0.5238095   0.8219178 0.7739464 22  39 180 20
## 20      0.24   0.5238095   0.8219178 0.7739464 22  39 180 20
## 21      0.25   0.5238095   0.8219178 0.7739464 22  39 180 20
## 22      0.26   0.5238095   0.8219178 0.7739464 22  39 180 20
## 23      0.27   0.4523810   0.8447489 0.7816092 19  34 185 23
## 24      0.28   0.4523810   0.8447489 0.7816092 19  34 185 23
## 25      0.29   0.4523810   0.8447489 0.7816092 19  34 185 23
## 26      0.30   0.4523810   0.8447489 0.7816092 19  34 185 23
## 27      0.31   0.3571429   0.8767123 0.7931034 15  27 192 27
## 28      0.32   0.3571429   0.8767123 0.7931034 15  27 192 27
## 29      0.33   0.3571429   0.8767123 0.7931034 15  27 192 27
## 30      0.34   0.3571429   0.8767123 0.7931034 15  27 192 27
## 31      0.35   0.3095238   0.9406393 0.8390805 13  13 206 29
## 32      0.36   0.3095238   0.9406393 0.8390805 13  13 206 29
```

```
## 33       0.37   0.3095238   0.9406393 0.8390805 13  13 206 29
## 34       0.38   0.3095238   0.9406393 0.8390805 13  13 206 29
## 35       0.39   0.3095238   0.9406393 0.8390805 13  13 206 29
## 36       0.40   0.2380952   0.9543379 0.8390805 10  10 209 32
## 37       0.41   0.2380952   0.9543379 0.8390805 10  10 209 32
## 38       0.42   0.2380952   0.9543379 0.8390805 10  10 209 32
## 39       0.43   0.2380952   0.9543379 0.8390805 10  10 209 32
## 40       0.44   0.1904762   0.9863014 0.8582375  8   3 216 34
## 41       0.45   0.1904762   0.9863014 0.8582375  8   3 216 34
## 42       0.46   0.1904762   0.9863014 0.8582375  8   3 216 34
## 43       0.47   0.1904762   0.9863014 0.8582375  8   3 216 34
## 44       0.48   0.1666667   0.9863014 0.8544061  7   3 216 35
## 45       0.49   0.1666667   0.9863014 0.8544061  7   3 216 35
## 46       0.50   0.1666667   0.9863014 0.8544061  7   3 216 35
## 47       0.51   0.1666667   0.9863014 0.8544061  7   3 216 35
## 48       0.52   0.1666667   0.9863014 0.8544061  7   3 216 35
## 49       0.53   0.1666667   0.9908676 0.8582375  7   2 217 35
## 50       0.54   0.1666667   0.9908676 0.8582375  7   2 217 35
## 51       0.55   0.1666667   0.9908676 0.8582375  7   2 217 35
## 52       0.56   0.1666667   0.9908676 0.8582375  7   2 217 35
## 53       0.57   0.1666667   0.9908676 0.8582375  7   2 217 35
## 54       0.58   0.1666667   0.9908676 0.8582375  7   2 217 35
## 55       0.59   0.1666667   0.9908676 0.8582375  7   2 217 35
## 56       0.60   0.1666667   0.9908676 0.8582375  7   2 217 35
```

```r
ggplot(knn_results, aes(x = threshold)) +
  geom_line(aes(y = Sensitivity, color = "Sensitivity"), linewidth = 1) +
  geom_line(aes(y = Specificity, color = "Specificity"), linewidth = 1) +
  geom_hline(yintercept = 0.60, linetype = "dashed") +
  labs(title = "Sensitivity & Specificity by Threshold",
       x = "Threshold",
       y = "Metric Value",
       color = "Metric") +
  theme_minimal()
```




```r
# This line of code calculates Balanced Accuracy
knn_results$BalancedAccuracy <-
  (knn_results$Sensitivity + knn_results$Specificity) / 2

knn_results[which.max(knn_results$BalancedAccuracy), ]
```

```
##    threshold Sensitivity Specificity  Accuracy TP FP  TN FN BalancedAccuracy
## 14      0.18   0.5714286   0.7853881 0.7509579 24 47 172 18        0.6784083
```

```r
knn_k23 <- train(
  Attrition_factor ~ .,
  data = trainData_knn2,
  method = "knn",
  preProcess = c("center", "scale"),
  tuneGrid = data.frame(k = 23)
)

prob_yes_knn <- predict(knn_k23,
                        newdata = testData_knn2,
                        type = "prob")[, "Yes"]

threshold <- 0.09
# Used the threshold found by Balanced Accuracy

pred_knn_adj <- factor(ifelse(prob_yes_knn >= threshold, "Yes", "No"),
                       levels = c("No", "Yes"))

kNNcm <- confusionMatrix(pred_knn_adj,
```

```
                testData_knn2$Attrition_factor,
                positive = "Yes")

print("kNN Confusion Matrix")
```

```
## [1] "kNN Confusion Matrix"
```

```
kNNcm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  144  15
##        Yes  75  27
##
##                Accuracy : 0.6552
##                  95% CI : (0.5941, 0.7127)
##     No Information Rate : 0.8391
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.1904
##
##  Mcnemar's Test P-Value : 4.999e-10
##
##             Sensitivity : 0.6429
##             Specificity : 0.6575
##          Pos Pred Value : 0.2647
##          Neg Pred Value : 0.9057
##              Prevalence : 0.1609
##          Detection Rate : 0.1034
##    Detection Prevalence : 0.3908
##       Balanced Accuracy : 0.6502
##
##        'Positive' Class : Yes
##
```

Now work on the Naive Bayes model for comparison

```
#Data wrangling - while not removing as many variables due to noise, I am removing variables that are c

employee_nb <- employee_data %>%
  select(
    Attrition,
    OverTime,
    JobLevel,
    JobInvolvement,
    MaritalStatus,
    TotalWorkingYears,
    WorkLifeBalance,
    MonthlyIncome,
    Age,
    BusinessTravel
  ) %>%
```

```r
  mutate(
    Attrition = factor(Attrition, levels = c("No","Yes")),
    OverTime = factor(OverTime),
    JobLevel = factor(JobLevel),
    JobInvolvement = factor(JobInvolvement),
    MaritalStatus = factor(MaritalStatus),
    WorkLifeBalance = factor(WorkLifeBalance),
    BusinessTravel = factor(BusinessTravel)
  ) %>%
  droplevels()

trainData_nb <- employee_nb[trainIndices, ]
testData_nb  <- employee_nb[-trainIndices, ]

nb_model <- naiveBayes(Attrition ~ ., data = trainData_nb, laplace = 1)

prob_yes <- predict(nb_model,
                    newdata = testData_nb,
                    type = "raw")[, "Yes"]

threshold <- 0.11
pred_nb_adj <- factor(ifelse(prob_yes >= threshold, "Yes", "No"),
                   levels = c("No","Yes"))

nbcm <- confusionMatrix(pred_nb_adj,
                testData_nb$Attrition,
                positive = "Yes")
print("Naive Bayes Confusion Matrix:")
```

```
## [1] "Naive Bayes Confusion Matrix:"
```

```r
nbcm
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  No Yes
##        No  133  15
##        Yes  86  27
##
##                Accuracy : 0.613
##                  95% CI : (0.551, 0.6724)
##     No Information Rate : 0.8391
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.1486
##
##  Mcnemar's Test P-Value : 3.278e-12
##
##             Sensitivity : 0.6429
##             Specificity : 0.6073
##          Pos Pred Value : 0.2389
##          Neg Pred Value : 0.8986
##              Prevalence : 0.1609
##          Detection Rate : 0.1034
```

```
##    Detection Prevalence : 0.4330
##       Balanced Accuracy : 0.6251
##
##        'Positive' Class : Yes
##
```

```
## kNN and Naive Bayes are very close
```

Calculate the cost for each model

```
retention_Incentive = 200.00

kNN_retention_Cost = retention_Incentive * (kNNcm$table[2,2] + kNNcm$table[2,1])
nb_retention_Cost = retention_Incentive * (nbcm$table[2,2] + nbcm$table[2,1])
print("kNN Retention Cost:")
```

```
## [1] "kNN Retention Cost:"
```

```
kNN_retention_Cost
```

```
## [1] 20400
```

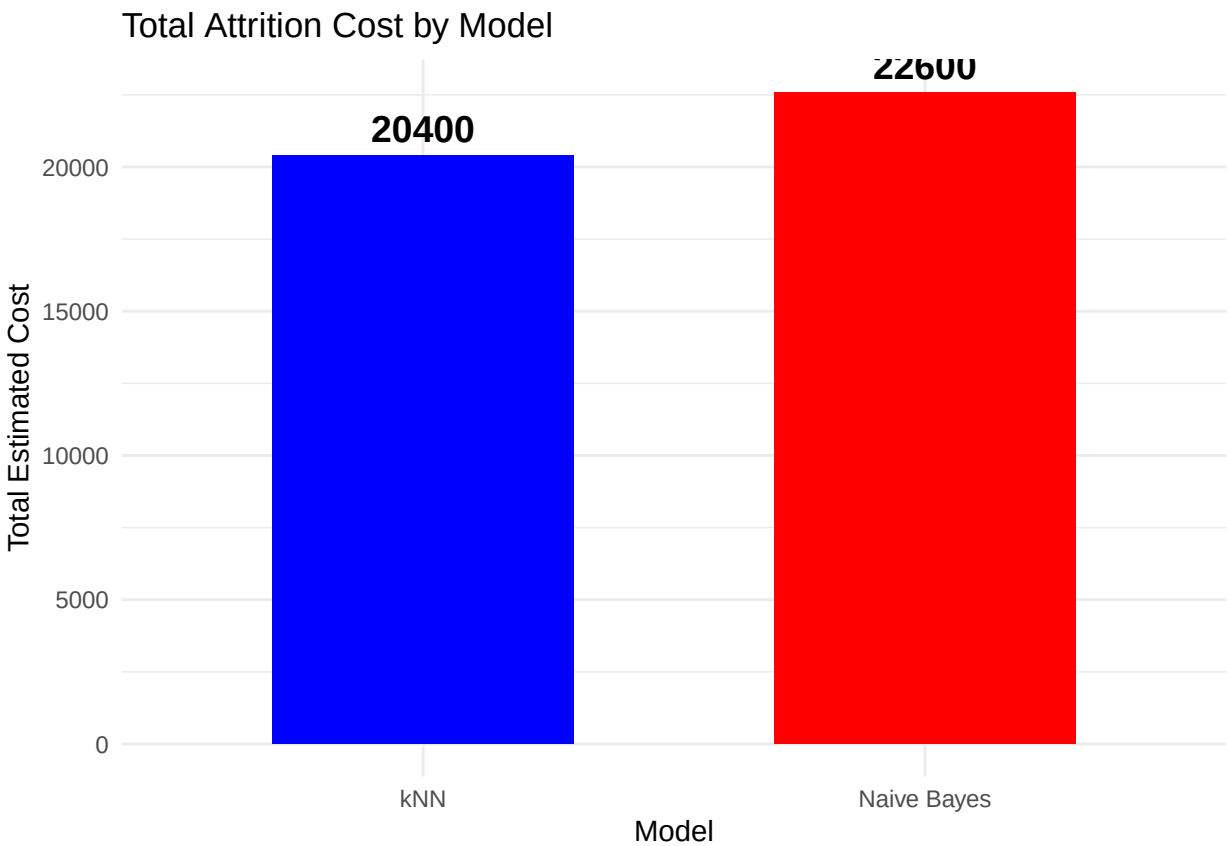
```
print("Naive Bayes Retention Cost:")
```

```
## [1] "Naive Bayes Retention Cost:"
```

```
nb_retention_Cost
```

```
## [1] 22600
```

```
cost_results <- data.frame(
  Model = c("kNN", "Naive Bayes"),
  TotalCost = c(kNN_retention_Cost, nb_retention_Cost)
)

ggplot(cost_results, aes(x = Model, y = TotalCost, fill = Model)) +
  geom_bar(stat = "identity", width = 0.6) +
  geom_text(aes(label = round(TotalCost, 0)),
            vjust = -0.5,
            size = 5,
            fontface = "bold") +
  scale_fill_manual(values = c("kNN" = "blue",
                               "Naive Bayes" = "red")) +
  labs(title = "Total Attrition Cost by Model",
       x = "Model",
       y = "Total Estimated Cost") +
  theme_minimal() +
  theme(legend.position = "none")
```




```
## This makes it appear that the kNN model is best. However, we haven't evaluated total savings yet.
```

Calculate total savings

A key component in the savings model is that it says that the cost to save an employee can be between 50-400% of their MonthlyIncome annualized.

Replacement cost assumptions were modeled between 0.5× and 4× annual salary, consistent with research from SHRM (2017) and Boushey & Glynn (2012), which estimate turnover costs ranging from 50% to over 200% of annual compensation depending on role complexity.

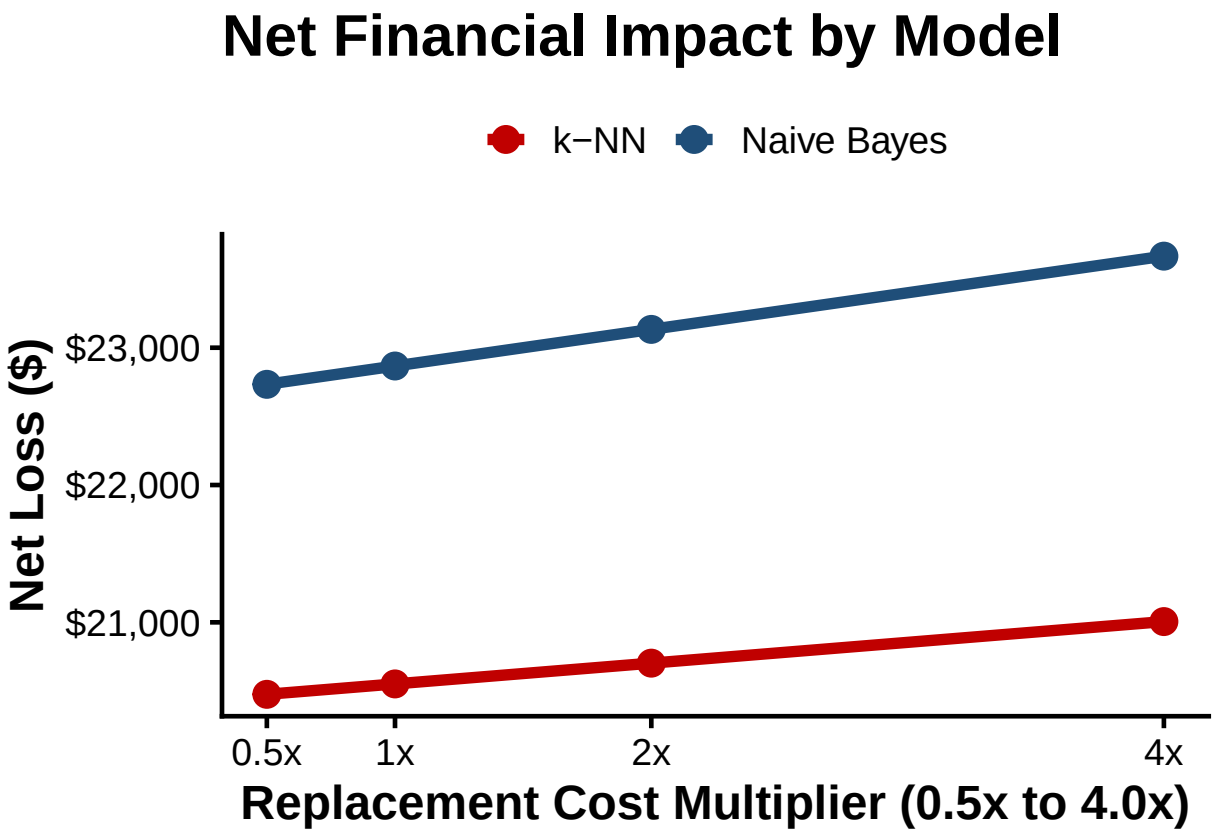
Boushey, H., & Glynn, S. J. (2012). There are significant business costs to replacing employees. Center for American Progress.

Society for Human Resource Management. (2017). Human Capital Benchmarking Report.I will calculate at .5x, 1x, 2x and 4x

```
trainData <- kNN_data2[trainIndices, ]
testData  <- kNN_data2[-trainIndices, ]

testData$AnnualIncome <- testData$MonthlyIncome * 12
```

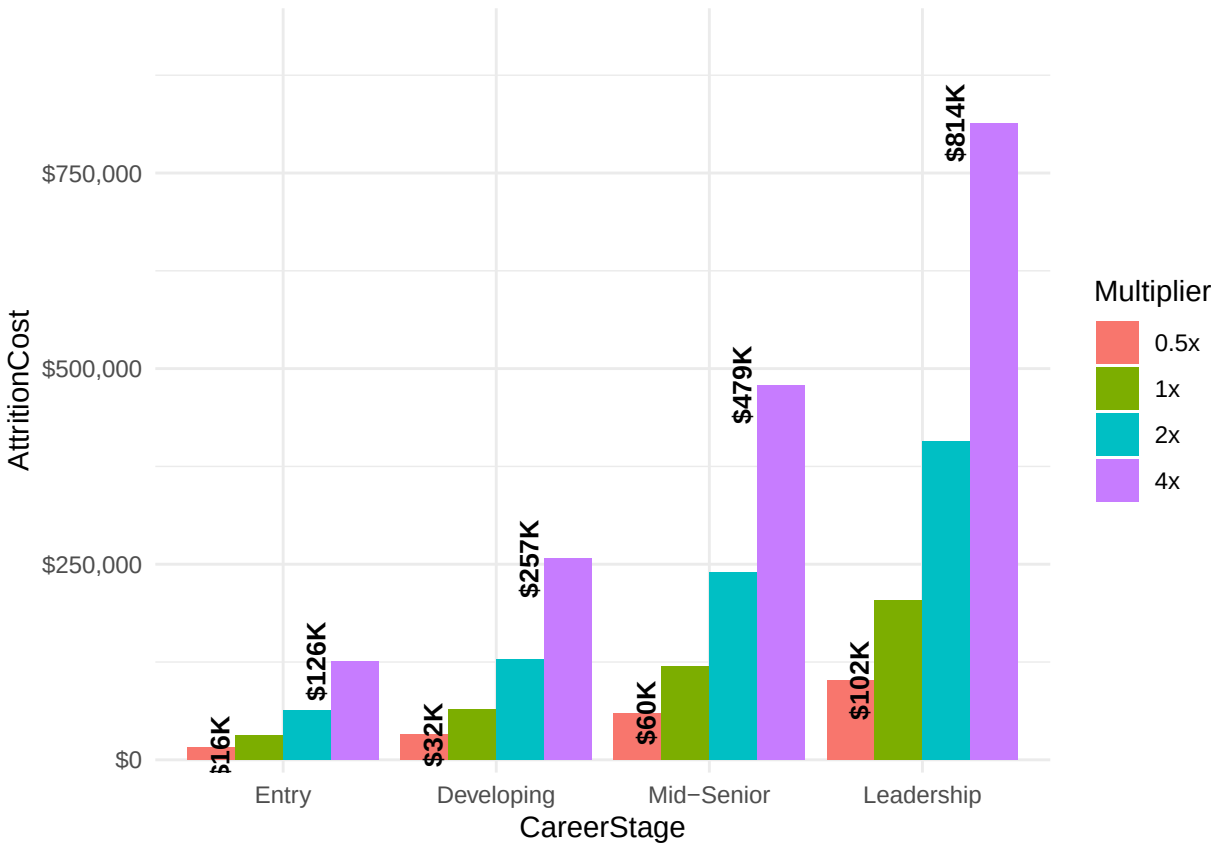
```r
calculate_savings <- function(predictions, actuals, income, multiplier) {

  TP_rows <- which(predictions == "Yes" & actuals == "Yes")
  FN_rows <- which(predictions == "No"  & actuals == "Yes")

  TP_income <- sum(income[TP_rows])
  FN_income <- sum(income[FN_rows])

  retention_cost <- sum(predictions == "Yes") * 200

  money_saved <- TP_income * multiplier
  replacement_cost <- FN_income * multiplier

  net_savings <- money_saved - retention_cost - replacement_cost

  return(net_savings)
}

multipliers <- c(0.5, 1, 2, 4)

knn_results <- sapply(multipliers, function(m) {
  calculate_savings(
    pred_knn_adj,
    testData$Attrition_factor,
    testData$AnnualIncome,
    m
  )
})

nb_results <- sapply(multipliers, function(m) {
  calculate_savings(
    pred_nb_adj,
    testData$Attrition,
    testData$MonthlyIncome * 12,
    m
  )
})

financial_results <- data.frame(
  Multiplier = multipliers,
  kNN_NetSavings = knn_results,
  NB_NetSavings = nb_results
)

financial_long <- financial_results %>%
  pivot_longer(
    cols = c(kNN_NetSavings, NB_NetSavings),
    names_to = "Model",
    values_to = "NetSavings"
  ) %>%
  mutate(
    NetLoss = -NetSavings,
    Model = recode(Model,
```

```r
                    kNN_NetSavings = "k-NN",
                    NB_NetSavings  = "Naive Bayes")
  )

ggplot(financial_long,
       aes(x = Multiplier,
           y = NetLoss,
           color = Model,
           group = Model)) +
  geom_line(linewidth = 2) +
  geom_point(size = 4) +
  scale_x_continuous(
    breaks = multipliers,
    labels = paste0(multipliers, "x")
  ) +
  scale_y_continuous(labels = dollar) +
  scale_color_manual(
    values = c(
      "Naive Bayes" = "#1F4E79",
      "k-NN"        = "#C00000"
    )
  ) +
  labs(
    title = "Net Financial Impact by Model",
    x = "Replacement Cost Multiplier (0.5x to 4.0x)",
    y = "Net Loss ($)"
  ) +
  theme_classic(base_size = 18) +
  theme(
    legend.title = element_blank(),
    plot.title = element_text(face = "bold"),
    axis.title = element_text(face = "bold"),
    legend.position = "top"
  )
```




```
## Across all replacement cost scenarios, from 0.5× to 4× annual salary, the Naive Bayes model consiste

## Now pair this data with the career stage information found earlier to get a view of the career stage

library(dplyr)
library(tidyr)
library(ggplot2)
library(scales)

employee_data_career <- employee_data_clean %>%
  mutate(
    CareerStage = case_when(
      JobLevel == 1 ~ "Entry",
      JobLevel == 2 ~ "Developing",
      JobLevel == 3 ~ "Mid-Senior",
      JobLevel >= 4 ~ "Leadership"
    )
  )

salary_stage <- employee_data_career %>%
  group_by(CareerStage) %>%
  summarise(
    MedianSalary = median(MonthlyIncome, na.rm = TRUE),
    .groups = "drop"
  )

multipliers <- tibble(
```

```r
  Multiplier = c(0.5, 1, 2, 4)
)

cost_table <- crossing(salary_stage, multipliers) %>%
  mutate(
    AttritionCost = MedianSalary * 12 * Multiplier,
    CareerStage = factor(CareerStage,
                         levels = c("Entry","Developing","Mid-Senior","Leadership")),
    Multiplier = factor(Multiplier,
                        levels = c(0.5,1,2,4),
                        labels = c("0.5x","1x","2x","4x"))
  )

# Create label data for only the lowest and highest scenarios
label_data <- cost_table %>%
  filter(Multiplier %in% c("0.5x","4x"))

ggplot(cost_table, aes(x = CareerStage, y = AttritionCost, fill = Multiplier)) +
  geom_col(position = "dodge") +
  geom_text(
  data = label_data,
  aes(label = label_dollar(scale = 1e-3, suffix = "K", accuracy = 1)(AttritionCost)),
  position = position_dodge(width = 0.9),
  angle = 90,
  vjust = -0.3,
  hjust = 0.5,
  size = 3.5,
  fontface = "bold"
) +
  scale_y_continuous(
  labels = dollar_format(),
  expand = expansion(mult = c(0.02, 0.18))
) +
  theme_minimal()
```



Produce an executive level summary

```
exec_table <- employee_data %>%
  count(Attrition) %>%
  mutate(
    Status = ifelse(Attrition == "No",
                    "Retained (Stayed)",
                    "Attrition (Lost)"),
    Percentage = percent(n / sum(n), accuracy = 0.1)
  ) %>%
  select(Status, Count = n, Percentage) %>%
  bind_rows(
    data.frame(
      Status = "Total Employees",
      Count = sum(.$Count),
      Percentage = "100%"
    )
  )


exec_table %>%
  kable(
    col.names = c("Status", "Count", "Percentage"),
    align = c("l", "c", "c")
  ) %>%
  kable_styling(
```

```r
    full_width = FALSE,
    position = "left",
    font_size = 16
  ) %>%
  row_spec(1, bold = TRUE) %>%
  row_spec(2,
           bold = TRUE,
           background = "#FDECEC",
           color = "#B22222") %>%
  row_spec(3, bold = TRUE) %>%
  column_spec(2, bold = TRUE) %>%
  column_spec(3, bold = TRUE)
```

Status	Count	Percentage
Retained (Stayed)	**730**	**83.9%**
Attrition (Lost)	**140**	**16.1%**
Total Employees	**870**	**100%**

```r
# Create a circular chart

attr_rate <- employee_data %>%
  summarise(rate = mean(Attrition == "Yes")) %>%
  pull(rate)

# Order matters: Stayed first, Attrited second
donut_df <- data.frame(
  category = c("Stayed", "Attrited"),
  value = c(1 - attr_rate, attr_rate)
)

ggplot(donut_df, aes(x = 2, y = value, fill = category)) +
  geom_col(color = "white", linewidth = 1.2) +

  # Rotate so orange lands bottom-left
  coord_polar(theta = "y", start = pi * 0.35) +

  xlim(0.5, 2.5) +  # controls hole size

  scale_fill_manual(values = c(
    "Stayed" = "#0B3A66",
    "Attrited" = "#D97706"
  )) +

  theme_void() +
  theme(legend.position = "none") +

  annotate("text", x = 0, y = 0,
           label = percent(attr_rate, accuracy = 0.1),
           size = 18, fontface = "bold") +
```

```
  annotate("text", x = 0, y = 0,
           label = "\nAttrition Rate",
           size = 6)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_text()`).
## Removed 1 row containing missing values or values outside the scale range
## (`geom_text()`).
```

Charts for presentation Purposes

```
## Create a data frame with the top 3 attrition risks: Stock Options, OverTime and Job Level


# ---- 1) Build the 5 segments (adjust thresholds if your EDA used different cutoffs) ----
risk_df <- bind_rows(

  # 1) Zero Stock Options
  employee_data_clean %>%
    filter(StockOptionLevel == 0) %>%
    summarise(RiskFactor = "Zero Stock Options",
              Left = sum(Attrition_factor == "Yes")),

  # 2) Job Level 1 only (Entry Level)
  employee_data_clean %>%
    filter(JobLevel == 1) %>%
    summarise(RiskFactor = "Job Level 1\n(Entry Level)",
              Left = sum(Attrition_factor == "Yes")),

  # 3) Mandatory OverTime
  employee_data_clean %>%
    filter(OverTime == "Yes") %>%
    summarise(RiskFactor = "Mandatory OverTime",
              Left = sum(Attrition_factor == "Yes")),
```

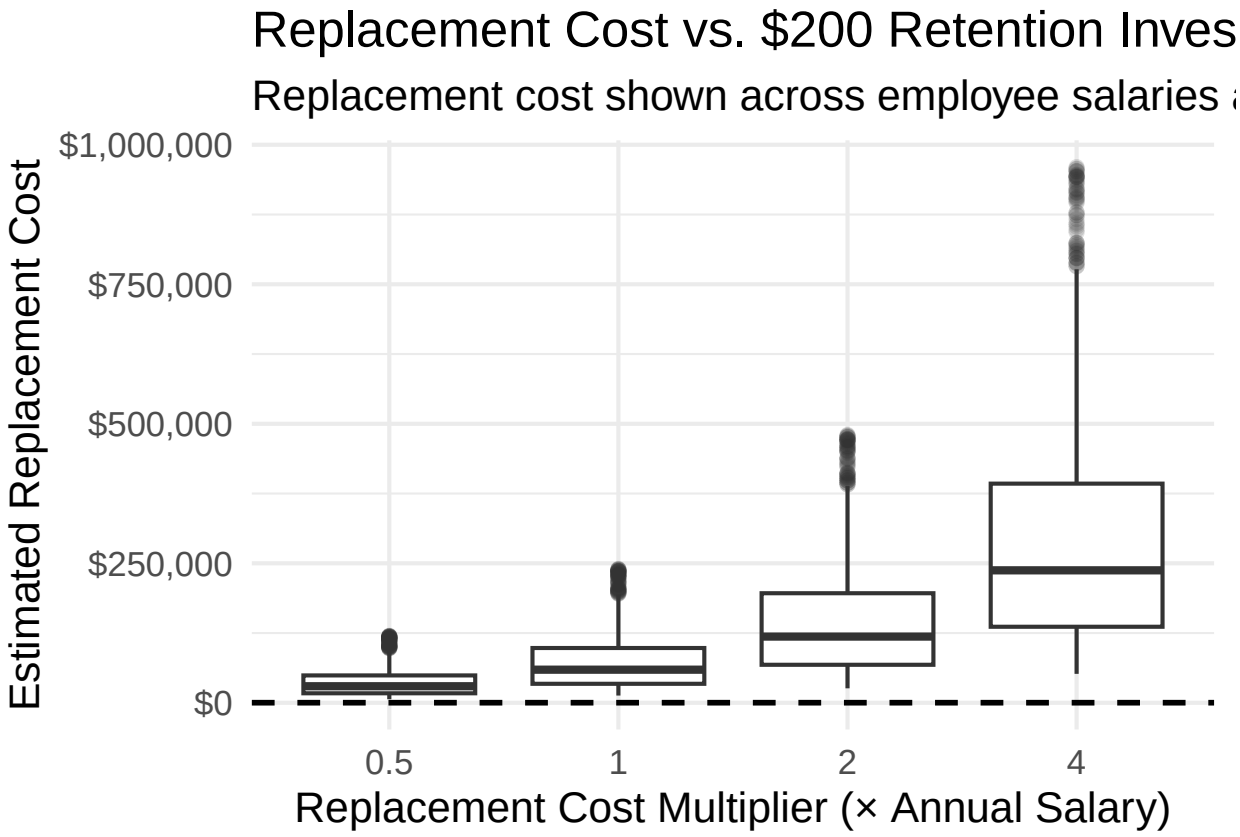
```r
  # 4) Low Environment Satisfaction (commonly 1 = low)
  employee_data_clean %>%
    filter(EnvironmentSatisfaction == 1) %>%
    summarise(RiskFactor = "Low Environment\nSatisfaction",
              Left = sum(Attrition_factor == "Yes")),

  # 5) Frequent Travel
  employee_data_clean %>%
    filter(BusinessTravel == "Travel_Frequently") %>%
    summarise(RiskFactor = "Frequent Travel",
              Left = sum(Attrition_factor == "Yes"))

) %>%
  mutate(
    # Top 3 = red; bottom 2 = gray (like the slide)
    Tier = ifelse(RiskFactor %in% c("Zero Stock Options",
                                    "Job Level 1\n(Entry Level)",
                                    "Mandatory OverTime"),
                  "Top", "Lower"),
    BarColor = ifelse(Tier == "Top", "#C81E1E", "#5A5A5A"),
    Label = ifelse(Tier == "Top",
                   paste0(Left, " Employees Lost"),
                   "Lower Volume")
  ) %>%
  arrange(desc(Left)) %>%
  mutate(RiskFactor = factor(RiskFactor, levels = rev(RiskFactor)))

# ---- 2) Plot (slide-style) ----
p <- ggplot(risk_df, aes(x = Left, y = RiskFactor)) +

  geom_col(fill = risk_df$BarColor, width = 0.65) +

  # Label inside top bars (white), "Lower Volume" outside gray bars (black)
  geom_text(aes(label = Label),
            color = ifelse(risk_df$Tier == "Top", "white", "black"),
            fontface = "bold",
            size = 5,
            hjust = ifelse(risk_df$Tier == "Top", 1, 0),
            nudge_x = ifelse(risk_df$Tier == "Top", -2, 2)) +

  labs(
    title = "Top Attrition Risk Segments",
    x = NULL,
    y = NULL
  ) +

  theme_minimal(base_size = 18) +
  theme(
    plot.title = element_text(face = "bold", size = 34, color = "black"),
    axis.text.y = element_text(size = 18, color = "black", face = "bold"),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    panel.grid = element_blank(),
```

```
    plot.background  = element_rect(fill = "#E8F0F4", color = NA),
    panel.background = element_rect(fill = "#E8F0F4", color = NA),
    plot.margin = margin(20, 40, 20, 40)
  ) +

  # Give extra room for "Employees Lost" text to the right
  expand_limits(x = max(risk_df$Left) * 1.25)

p
```

Show the difference between cost and true cost

```
# 1) Build annual salary + scenario costs
cost_df <- employee_data_clean %>%
  mutate(AnnualSalary = MonthlyIncome * 12) %>%
  crossing(multiplier = c(0.5, 1, 2, 4)) %>%
  mutate(ReplacementCost = AnnualSalary * multiplier,
         IncentiveCost = 200)

# 2) Summarize for boxplot look (distribution)
ggplot(cost_df, aes(x = factor(multiplier), y = ReplacementCost)) +
  geom_boxplot(outlier.alpha = 0.15) +
  geom_hline(yintercept = 200, linetype = "dashed", linewidth = 1) +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "Replacement Cost vs. $200 Retention Investment",
    subtitle = "Replacement cost shown across employee salaries at 0.5×-4× annual salary",
```

```
    x = "Replacement Cost Multiplier (× Annual Salary)",
    y = "Estimated Replacement Cost"
  ) +
  theme_minimal(base_size = 16)
```




```
median_salary <- employee_data_clean %>%
  mutate(AnnualSalary = MonthlyIncome * 12) %>%
  summarise(med_salary = median(AnnualSalary, na.rm = TRUE)) %>%
  pull(med_salary)

median_table <- data.frame(
  Scenario = c("0.5× Salary", "1× Salary", "2× Salary", "4× Salary"),
  `Replacement Cost` = dollar(median_salary * c(0.5, 1, 2, 4))
)

median_table %>%
  kable(
    align = "lc",
    caption = "Estimated Replacement Cost (Median Salary)"
  ) %>%
  kable_styling(
    full_width = FALSE,
    font_size = 20,
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE) %>%
  row_spec(1, bold = TRUE, background = "#FDECEC")
```

Table 17: Estimated Replacement Cost (Median Salary)

Scenario	Replacement.Cost
0.5× Salary	**$29,673**
1× Salary	$59,346
2× Salary	$118,692
4× Salary	$237,384

Table ready confusion matrix

```
cm_scorecard <- function(cm_obj, model_name = "Model") {
  tab <- cm_obj$table
  TP <- tab["Yes","Yes"]
  TN <- tab["No","No"]
  FP <- tab["Yes","No"]
  FN <- tab["No","Yes"]

  tibble(
    Model = model_name,
    TP = TP, FP = FP, TN = TN, FN = FN,
    Accuracy = percent(unname(cm_obj$overall["Accuracy"]), 0.1),
    Sensitivity = percent(unname(cm_obj$byClass["Sensitivity"]), 0.1),
    Specificity = percent(unname(cm_obj$byClass["Specificity"]), 0.1),
    Precision = percent(unname(cm_obj$byClass["Pos Pred Value"]), 0.1)
  )
}

score_tbl <- bind_rows(
  cm_scorecard(kNNcm, "k-NN (k=23, t=0.09)"),
  cm_scorecard(nbcm,  "Naive Bayes (t=0.11)")
)

kable(score_tbl, caption = "Model Confusion Matrix Scorecard") %>%
  kable_styling(full_width = FALSE, font_size = 18) %>%
  row_spec(0, bold = TRUE) %>%
  column_spec(1, bold = TRUE)
```

Table 18: Model Confusion Matrix Scorecard

Model	TP	FP	TN	FN	Accuracy	Sensitivit

k-NN (k=23, t=0.09)	27	75	144	15	65.5%	64.3%
Naive Bayes (t=0.11)	27	86	133	15	61.3%	64.3%

Competition output for new csv file

```r
comp_data = read.csv("/Users/dianewroberts/Case Study 1/CaseStudy1CompSet No Attrition.csv", sep = ",",

head(comp_data)
```

```
##     ID Age    BusinessTravel DailyRate             Department DistanceFromHome
## 1 1171  35     Travel_Rarely       750 Research & Development               28
## 2 1172  33     Travel_Rarely       147        Human Resources                2
## 3 1173  26     Travel_Rarely      1330 Research & Development               21
## 4 1174  55     Travel_Rarely      1311 Research & Development                2
## 5 1175  29     Travel_Rarely      1246                  Sales               19
## 6 1176  51 Travel_Frequently      1456 Research & Development                1
##   Education  EducationField EmployeeCount EmployeeNumber
## 1         3   Life Sciences             1           1596
## 2         3 Human Resources             1           1207
## 3         3         Medical             1           1107
## 4         3   Life Sciences             1            505
## 5         3   Life Sciences             1           1497
## 6         4         Medical             1            145
##   EnvironmentSatisfaction Gender HourlyRate JobInvolvement JobLevel
## 1                       2   Male         46              4        2
## 2                       2   Male         99              3        1
## 3                       1   Male         37              3        1
## 4                       3 Female         97              3        4
## 5                       3   Male         77              2        2
## 6                       1 Female         30              2        3
##                     JobRole JobSatisfaction MaritalStatus MonthlyIncome
## 1     Laboratory Technician               3       Married          3407
## 2           Human Resources               3       Married          3600
## 3     Laboratory Technician               3      Divorced          2377
## 4                   Manager               4        Single         16659
## 5           Sales Executive               3      Divorced          8620
## 6 Healthcare Representative               1        Single          7484
##   MonthlyRate NumCompaniesWorked Over18 OverTime PercentSalaryHike
## 1       25348                  1      Y       No                17
## 2        8429                  1      Y       No                13
## 3       19373                  1      Y       No                20
## 4       23258                  2      Y      Yes                13
## 5       23757                  1      Y       No                14
## 6       25796                  3      Y       No                20
##   PerformanceRating RelationshipSatisfaction StandardHours StockOptionLevel
## 1                 3                        4            80                2
## 2                 3                        4            80                1
## 3                 4                        3            80                1
## 4                 3                        3            80                0
## 5                 3                        3            80                2
## 6                 4                        3            80                0
##   TotalWorkingYears TrainingTimesLastYear WorkLifeBalance YearsAtCompany
```

```
## 1                10                     3               2             10
## 2                 5                     2               3              5
## 3                 1                     0               2              1
## 4                30                     2               3              5
## 5                10                     3               3             10
## 6                23                     1               2             13
##   YearsInCurrentRole YearsSinceLastPromotion YearsWithCurrManager
## 1                  9                       6                    8
## 2                  4                       1                    4
## 3                  1                       0                    0
## 4                  4                       1                    2
## 5                  7                       0                    4
## 6                 12                      12                    8
```

```r
str(comp_data)
```

```
## 'data.frame':    300 obs. of  35 variables:
##  $ ID                      : int  1171 1172 1173 1174 1175 1176 1177 1178 1179 1180 ...
##  $ Age                     : int  35 33 26 55 29 51 52 39 31 31 ...
##  $ BusinessTravel          : chr  "Travel_Rarely" "Travel_Rarely" "Travel_Rarely" "Travel_Rarely" ..
##  $ DailyRate               : int  750 147 1330 1311 1246 1456 585 1387 1062 534 ...
##  $ Department              : chr  "Research & Development" "Human Resources" "Research & Development"
##  $ DistanceFromHome        : int  28 2 21 2 19 1 29 10 24 20 ...
##  $ Education               : int  3 3 3 3 3 4 4 5 3 3 ...
##  $ EducationField          : chr  "Life Sciences" "Human Resources" "Medical" "Life Sciences" ...
##  $ EmployeeCount           : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ EmployeeNumber          : int  1596 1207 1107 505 1497 145 2019 1618 1252 587 ...
##  $ EnvironmentSatisfaction : int  2 2 1 3 3 1 1 2 3 1 ...
##  $ Gender                  : chr  "Male" "Male" "Male" "Female" ...
##  $ HourlyRate              : int  46 99 37 97 77 30 40 76 96 66 ...
##  $ JobInvolvement          : int  4 3 3 3 2 2 3 3 2 3 ...
##  $ JobLevel                : int  2 1 1 4 2 3 1 2 2 3 ...
##  $ JobRole                 : chr  "Laboratory Technician" "Human Resources" "Laboratory Technician"
##  $ JobSatisfaction         : int  3 3 3 4 3 1 4 1 1 3 ...
##  $ MaritalStatus           : chr  "Married" "Married" "Divorced" "Single" ...
##  $ MonthlyIncome           : int  3407 3600 2377 16659 8620 7484 3482 5377 6812 9824 ...
##  $ MonthlyRate             : int  25348 8429 19373 23258 23757 25796 19788 3835 17198 22908 ...
##  $ NumCompaniesWorked      : int  1 1 1 2 1 3 2 2 1 3 ...
##  $ Over18                  : chr  "Y" "Y" "Y" "Y" ...
##  $ OverTime                : chr  "No" "No" "No" "Yes" ...
##  $ PercentSalaryHike       : int  17 13 20 13 14 20 15 13 19 12 ...
##  $ PerformanceRating       : int  3 3 4 3 3 4 3 3 3 3 ...
##  $ RelationshipSatisfaction: int  4 4 3 3 3 3 2 4 2 1 ...
##  $ StandardHours           : int  80 80 80 80 80 80 80 80 80 80 ...
##  $ StockOptionLevel        : int  2 1 1 0 2 0 2 3 0 0 ...
##  $ TotalWorkingYears       : int  10 5 1 30 10 23 16 10 10 12 ...
##  $ TrainingTimesLastYear   : int  3 2 0 2 3 1 3 3 2 2 ...
##  $ WorkLifeBalance         : int  2 3 2 3 3 2 2 3 3 3 ...
##  $ YearsAtCompany          : int  10 5 1 5 10 13 9 7 10 1 ...
##  $ YearsInCurrentRole      : int  9 4 1 4 7 12 8 7 9 0 ...
##  $ YearsSinceLastPromotion : int  6 1 0 1 0 12 0 7 1 0 ...
##  $ YearsWithCurrManager    : int  8 4 0 2 4 8 0 7 8 0 ...
```

```r
names(comp_data)
```

```
##  [1] "ID"                       "Age"
##  [3] "BusinessTravel"           "DailyRate"
##  [5] "Department"               "DistanceFromHome"
##  [7] "Education"                "EducationField"
##  [9] "EmployeeCount"            "EmployeeNumber"
## [11] "EnvironmentSatisfaction"  "Gender"
## [13] "HourlyRate"               "JobInvolvement"
## [15] "JobLevel"                 "JobRole"
## [17] "JobSatisfaction"          "MaritalStatus"
## [19] "MonthlyIncome"            "MonthlyRate"
## [21] "NumCompaniesWorked"       "Over18"
## [23] "OverTime"                 "PercentSalaryHike"
## [25] "PerformanceRating"        "RelationshipSatisfaction"
## [27] "StandardHours"            "StockOptionLevel"
## [29] "TotalWorkingYears"        "TrainingTimesLastYear"
## [31] "WorkLifeBalance"          "YearsAtCompany"
## [33] "YearsInCurrentRole"       "YearsSinceLastPromotion"
## [35] "YearsWithCurrManager"
```

```r
# Retrain the knn model

kNN_small_data2 <- employee_data_clean %>%
  mutate(
    Attrition_factor = factor(Attrition_factor, levels = c("No", "Yes")),
    OverTime_Yes = ifelse(OverTime == "Yes", 1, 0)
  ) %>%
  select(
    Attrition_factor,
    StockOptionLevel,
    JobLevel,
    OverTime_Yes,
    MonthlyIncome,
    Age
  )

set.seed(123)
trainIndices <- createDataPartition(
  kNN_small_data2$Attrition_factor,
  p = 0.70,
  list = FALSE
)

trainData_knn3 <- kNN_small_data2[trainIndices, ]
testData_knn3  <- kNN_small_data2[-trainIndices, ]

knn_k23b <- train(
  Attrition_factor ~ .,
  data = trainData_knn3,
  method = "knn",
  preProcess = c("center", "scale"),
  tuneGrid = data.frame(k = 23)
)
```

```r
library(dplyr)

comp_data_clean <- comp_data %>%
  mutate(
    OverTime_Yes = ifelse(OverTime == "Yes", 1, 0),
    MaritalStatus_Single = ifelse(MaritalStatus == "Single", 1, 0),
    Travel_Frequently = ifelse(BusinessTravel == "Travel_Frequently", 1, 0)
  ) %>%
  select(
    ID, # Keep ID for the final output order
    TotalWorkingYears,
    MonthlyIncome,
    Age,
    StockOptionLevel,
    JobSatisfaction,
    OverTime_Yes,
    MaritalStatus_Single,
    Travel_Frequently,
    JobInvolvement,
    WorkLifeBalance,
    JobLevel
  )

comp_data_clean_scaled <- comp_data_clean %>%
  mutate(across(-ID, scale))

prob_yes_comp <- predict(knn_k23b, newdata = comp_data_clean, type = "prob")[, "Yes"]
comp_predictions <- ifelse(prob_yes_comp >= 0.09, "Yes", "No")

submission_df <- data.frame(
  ID = comp_data_clean$ID,
  Attrition = comp_predictions
)

submission_df <- submission_df[order(submission_df$ID), ]

write.csv(submission_df, "/Users/dianewroberts/Case Study 1/Case1PredictionsRoberts Attrition.csv", row
```